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Automated Lumbar Spine Degenerative Classification Using Deep Learning: A Comprehensive Evaluation Based on RSNA 2024

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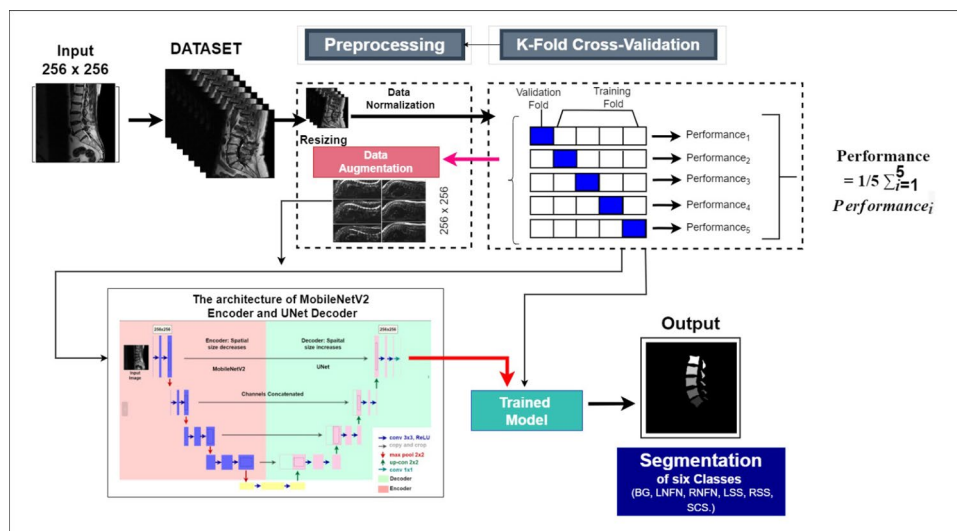
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Abstract

Degenerative lumbar spine diseases, including disk herniation, spinal stenosis, and facet arthropathy, are leading causes of chronic lower back pain, significantly affecting patients worldwide. Diagnosing these conditions through MRI imaging is time-consuming and subjective. This study introduces a novel deep-learning model using the MobileNetV2-UNet architecture to automate the classification and segmentation of lumbar spine degenerative diseases. The MobileNetV2-UNet model integrates MobileNetV2 for efficient feature extraction and UNet for accurate segmentation. The model was trained and validated on the RSNA 2024 Lumbar Spine Degenerative Classification dataset, which includes labeled MRI images. Depthwise separable convolutions enhanced computational efficiency, while skip connections in the UNet architecture preserved spatial details. The combined Cross-Entropy and IoU loss functions optimized classification accuracy and segmentation quality. The model's performance was evaluated using 5-fold cross-validation. The MobileNetV2-UNet model achieved a 94.93% accuracy, a 94.61% Dice Coefficient, and a 95.65% F1 score in classifying and segmenting various lumbar spine conditions, such as neural foraminal narrowing and spinal canal stenosis. The model's efficient architecture enables real-time application in clinical settings while maintaining high performance across diverse datasets. The proposed MobileNetV2-UNet model offers a reliable, efficient solution for automating lumbar spine degenerative disease diagnosis, reducing manual workload, and improving diagnostic precision. Future work will focus on enhancing generalization, improving model interpretability, and applying the framework to other musculoskeletal and neurological conditions, making it a promising tool for clinical use.



Graphical Abstract



Keywords MobileNetV2-UNet · Deep learning · Classification · Object detection · Medical imaging · Large dataset · Lumbar spine · Neural foraminal narrowing · Malignant · Spinal canal stenosis

1 Introduction

The World Health Organization predicts that degenerative changes (DC) in the lumbar spine frequently cause chronic lower back pain, impact living standards, and increase healthcare costs worldwide [1, 2]. A major factor underlying these problems is how to classify such degenerative changes as disk herniation, spinal stenosis, and facet arthropathy. This task depends mainly on contemporary medical imaging techniques like magnetic resonance imaging (MRI) and computed tomography (CT). However, this process is often subjective, time-consuming, and prone to individual variation in interpretation. As a result, findings may vary somewhat between readers of the same data set, which robs diagnosis of all certainty [1–3].

Lumbar spine degenerative diseases affect many people around the world, including older adults. Research shows that lumbar spinal stenosis leads to more spinal surgeries in people over 65 than any other condition. Each year, this problem impacts 250,000 to 500,000 people in the United States [4]. Imaging studies reveal that more than 60% of individuals over 60 have age-related degeneration in their lumbar spine, but not everyone experiences symptoms [3]. From a financial point of view, lumbar spine disorders place a heavy economic burden on society, including direct healthcare costs and lost productivity. In 2016, the U.S. spent \$134.5 billion to treat low back and neck pain. This made it the third most expensive health issue, just behind diabetes and heart disease [5]. The number of people with lumbar spine degenerative diseases will go up as people around the world get older. By 2050, the number of people 60 years and older worldwide is set to double to 2.1 billion [6]. This means more people will suffer from problems in their lower back, like stenosis, narrowing of nerve openings, and issues under the joints. By 2030, over 30% of people in many rich countries will have some spine problems [7].

Treatments that do not need surgery for spine problems are becoming more popular. Procedures that do not require significant cuts and treatments that help the body heal itself, like using stem cells and devices that space out the bones, are expected to grow a lot. The worldwide market for spine technologies that do not fuse bones is predicted to grow by 7.5% each year [8]. To overcome this problem, AI has made a significant breakthrough. Bringing AI into spinal diagnostics is set to cause a revolution in spotting and planning treatment for lumbar spine

issues early on. By 2035, experts think AI-powered image analysis will cut diagnostic times in half and boost accuracy in finding degenerative conditions in their early stages [9].

With the emergence of large-scale medical imaging datasets, deep learning techniques are increasingly popular in recognizing lumbar spine degenerative diseases [4–6]. Deep learning in medical imaging has shown promise when applied to large datasets. It has the ability to learn complex patterns in such data far better than conventional approaches and can outcompete it [3, 7]. Nevertheless, this tremendous potential has been limited by the fact that the existing deep learning methods are usually specific to a single individual or a medical facility. This renders it difficult to reproduce them elsewhere using data sets such as the quality and diversity that exist in practice [8, 9].

This article analyses the problem of lumbar spine degeneration classification by AIs. The data was provided by the Lumbar Spine Degenerative Classification Kaggle competition of 2024 of RSNA. This study is aimed at helping improve the existing methods of lumbar spine illness diagnosis and severity. The technique proposed saves time on the part of radiologists and enables physicians to attend to patients [6, 8, 10]. Next, another aspect of the study is the division of the affected area and the evaluation of severity. The AI-based lumbar degeneration categorization system can provide the correct results and valuable insights [11, 12]. The project will confirm the model with the help of the RSNA 2024 Lumbar Spine Degenerative Classification database and develop automated solutions based on the improved deep learning (DL) techniques. State-of-the-art AI lumbar spine degeneration classification systems enhance precise and rapid disease classification and segmentation across multiple types [4–6]. The study introduces three novelties to deep learning: The model has a lightweight MobileNetV2 backbone that helps extract features efficiently, combines CE + IoU loss with sensitivity analysis to balance between pixel-level and region-level performance, and a systematic ablation to display component contributions. These modifications make our study unique from the prior literature and enhance strength and performance. This paper confirms the MobileNetV2-UNet model on the large-scale RSNA 2024 dataset, although it requires external validation on multiple hospital datasets to demonstrate its applicability and reliability in clinical practice.

The key objectives of this study are:

1. **Deployment of MobileNetV2-UNet Architecture:** The study employs the MobileNetV2-UNet structure that incorporates the performance and size of MobileNetV2 and the segmentation power of UNet. It is designed so as to divide medical images with high accuracy and high computational efficiency, thus applicable in routine analysis and in high volume operation, such as in degenerative disease of the lumbar spine classification.
2. **Application to a Real-World Dataset with High-Resolution Segmentation and Classification:** It is trained on RSNA 2024 Lumbar Spine Degenerative Classification, which is a Kaggle competition dataset with numerous MRI images being differentially labeled. This data can be thoroughly assessed and checked with the help of a MobileNetV2-UNet model, which makes sure that our results will probably be applicable to clinical scenarios. In the second stage, we applied our MobileNetV2-UNet to classify and divide the degenerative changes of the lumbar spine, which include disk herniation, spinal stenosis, and facet arthropathy. Without a high-resolution output, the design is not able to accurately diagnose medical conditions.
3. **Efficiency in Clinical Application:** MobileNet V2 is built to work well and efficiently on resource-limited edge hardware (like a mobile phone or tablet) or low-power medical imaging equipment without loss in quality. This can be of some considerable use in clinical deployment because the computational burden can be significantly lowered, and subject-specific applications are available in real time.
4. **Generalization Across Diverse Data with Enhanced Interpretability and Visualization:** This paper trains and evaluates the MobileNetV2-UNet on a complete set of diverse subsets of the RSNA 2024 data to address generalization concerns: patient demographics (age, sex); medical device type-related imaging conditions among hospitals; and acquisition protocol changes over time. An emphasis on generalizability is crucial for creating a strong model applicable to a wide range of clinical cases. Advanced interpretability approaches like Grad-CAM are used to understand MobileNetV2-UNet categorization decision processes. Building trust with clinicians requires interpretability, so model predictions are precise and clinically sensitive.

5. Clinical Decision Support Integration aligned with Scalability and Broader Applications: Our study has demonstrated how the MobileNetV2-UNet model could be integrated into a clinical decision support system and provided a practical, real-time solution to the radiologists by analyzing degenerative changes in the lumbar spine. The purpose of this integration was to assist with increasing diagnostic certainty and reducing the amount of time required to perform manual image analysis. As much as we focus on lumbar spine degenerative alterations in our work, we also think that the MobileNetV2-UNet architecture and methodology can be generalized extensively to other medical imaging applications. The usage of this model in other musculoskeletal or neurological disorders is also addressed, and the scalability of such a study is critical in extensive implications on healthcare.

This is how the rest of the article is structured. Section 2 delves into a comprehensive discussion of the current methods in depth, followed by the proposed methodology in Sect. 3. The findings and analysis are detailed in Sect. 4. The references are followed by Sect. 5, which discusses the conclusion and future work.

2 Literature Work

The lumbar spine consists of L1 to L5, which support the upper body between the thoracic and sacrum. Lumbar vertebrae absorb axial pressure of the head, neck, and trunk and are larger than the other vertebrae on the spine. Three longitudinal arteries supply the spinal cord with arterial blood: an anterior (in front) vessel that runs along the spinal cord and two posterior spinal vessels that run on each side. It is cushioned and moves down the lumbar spine. They supply oxygenated blood to the anterior two-thirds and the posterior one-third of the spinal cord. Anterior and posterior radicular arteries collateralise the vertebrae. The quality of life can be permanently affected by degenerative changes (DC), including disk herniation, spinal stenosis, and facet arthropathy, because the L-spine is so functional. The most common causes of chronic lower back pain are not always associated with simpler imaging, such as an MRI scan or CT. Even though very decisive, the traditional diagnostic procedure of physicians is lengthy and has been subject to the experience or ability of a radiologist, and leads to varied interpretations of the reads. The emergence of medical imaging and artificial intelligence (AI) technologies has created intense learning applications with immense potential to automate the classification of degenerative changes of the lumbar spine and lessen the uncertainty in clinical diagnosis, and positively influence patient outcomes.

2.1 Degenerative Changes in the Lumbar Spine

The lumbar spine conditions include disk herniation, spinal stenosis, and facet arthropathy. These degenerative changes are leading contributors to chronic lower back pain, significantly influencing quality of life and healthcare costs worldwide [13, 14]. Diagnosis typically relies on MRI or CT imaging, but these methods are subjective, time-consuming, and dependent on radiologist expertise [15, 16]. Variability in interpretation between clinicians highlights the need for automated systems that can deliver consistent results.

2.2 The Role of Deep Learning in Medical Imaging

In recent years, deep learning has emerged as a powerful approach for diagnosing spinal degenerative diseases through medical imaging [4, 17]. Models trained on large datasets have shown strong potential to reduce diagnostic uncertainty and improve efficiency. However, these models often exhibit poor generalizability when applied to data outside their training distribution [8, 9]. Recent studies have explored domain adaptation techniques to mitigate this issue. For instance, Pan et al. [18] demonstrated that deep learning-based domain adaptation can reduce performance degradation across biomedical imaging datasets, underscoring the importance of robust cross-domain generalization in clinical practice. In addition to CNNs, Transformer-based vision Transformer

(ViT) and Swin Transformer have quickly become used in medical imaging in classification and segmentation. Recent studies represent a synthesis of over a hundred studies, which point to the fact that Transformers is characterized by a high degree of representational capacity and cross-domain transferability in medical tasks [19]. Such gains are usually, though not always, connected to greater compute and memory footprint as compared to lightweight CNN backbones, and this encourages us to listen to MobileNetV2 to be useful in clinical use.

2.3 Challenges in Deep Learning for Lumbar Spine Classification

Deep learning has a big challenge in the classification of the lumbar spine. It needs quality datasets of actual clinical instances or scenarios, to this end. The current literature demonstrates that deep learning models can be used to obtain high performance in specific data sets. But when they are trained on other sources of data or on the quality of a different image, they perform badly [12, 20]. This underperformance in various data is a critical problem and must be solved. It needs to be solved to implement these models in the clinics. However, one more problem is the understanding of deep learning models. These are efficient; however, doctors struggle to comprehend why these models make such forecasts. This black box problem may slow down the process of implementing such tools in hospitals and clinics [21, 22]. A lot of work is still required in order to onboard more doctors, which involves simplifying these models. This is significant in order that healthcare can be more open and accommodating to them.

2.4 Advances in Automated Classification Systems

There are a number of groups that have explored AI systems in line with lumbar spine disease detection. Ishimoto et al. [9] validated an AI model for MRI-based grading of lumbar spinal stenosis, while Saravi et al. [10] developed a segmentation-based system for quantifying dural sac areas. Siddique et al. [12] explored U-Net variants for spine imaging tasks, highlighting the usefulness of encoder–decoder structures. More recently, Yu et al. [23] introduced a multi-scale hybrid deep learning framework for spinal image classification, achieving improved diagnostic accuracy and demonstrating the value of architectural innovation in clinical contexts.

Others are also classification models, which can categorize and sort spinal problems with increased accuracy. Hallinan et al. [11] thus developed a deep learning model that identifies and categorizes central canal lateral recess and neural foraminal narrowing in lower back MRIs to be valuable in clinics and reduce the radiologists' time. Methodologically, Swin-style hierarchical Transformers have demonstrated competitive accuracy via shifted windows and local self-attention; e.g., SwinSight achieves strong results in large-scale visual tasks and continues to influence medical model design [29]. In spine/musculoskeletal imaging specifically, researchers have begun to tailor Transformer encoders—e.g., vertebrae-aware ViT variants for spine segmentation and Swin-based models for neuro-anatomical delineation—showcasing the promise and cost profile of Transformer backbones in these domains [30].

2.5 Emerging Techniques and Applications

New directions in deep learning for spine imaging focus on improving robustness and clinical interpretability. Advanced reconstruction approaches have been applied to accelerate MRI without losing diagnostic quality [26, 31, 32]. Landmark detection algorithms have also been proposed to support spinal alignment assessments [24, 25]. Beyond accuracy, researchers are now emphasizing explainability to improve trust in AI. For example, Zhang et al. [33] presented explainable AI methods tailored for musculoskeletal imaging, showing how attention mechanisms and saliency maps can enhance transparency and support clinical decision-making.

Existing studies have reported promising results in spinal or musculoskeletal imaging tasks, employing architectures such as ResNet, DenseNet, or Transformer-based models. While effective, these models often require heavy computation, making them less suitable for scalable clinical use. By contrast, our MobileNetV2-based

approach achieves competitive performance with significantly reduced complexity. Furthermore, unlike most prior studies, we provide an ablation analysis to experimentally demonstrate the contribution of each proposed component.

2.6 Limitations and Future Directions

Thus, even with these improvements, particular challenges are inherent to introducing deep learning into health-care. One primary concern regards the fact that deep learning models might not be sufficiently stable, as Antun and others [20] explained regarding the risks that AI presents regarding the restoration of images. These shaky models could lead to wrong diagnoses or miss something that could endanger the patient's life. Moreover, it means that measurements can differ; for instance, the Cobb angle used to ensure scoliosis is present [34]. For these new tools to be used, they must be functional in different types of scans and with various groups of patients. A key limitation of this study is its reliance on the RSNA 2024 competition dataset, which, while extensive and well-annotated, may not fully capture the heterogeneity of real-world hospital imaging, where scanner vendors, acquisition protocols, and patient populations vary widely. External validation on multi-institutional clinical datasets will therefore be essential to confirm the robustness and generalizability of the model. In addition, the combined CE + IoU loss function was evaluated only at discrete α values (0.3, 0.5, and 0.7). Although $\alpha = 0.5$ yielded the most stable performance, further systematic exploration across a broader range of values and on more diverse datasets could help optimize this parameter for different clinical contexts. Another limitation is that the present framework focuses on condition identification without explicit severity grading (e.g., mild, moderate, severe). Since clinical guidelines often rely on such gradings for therapeutic decisions, future datasets annotated with severity scales will be necessary to extend our model accordingly. A major limitation of this study, consistent with many prior works, is the reliance on a single-source competition dataset. Although the RSNA 2024 dataset offers high-quality annotations, it does not capture variations across hospitals, scanner vendors, and acquisition protocols. Testing on independent datasets not part of training or validation will be essential to demonstrate generalization. Future work will therefore focus on multi-institutional validation and domain adaptation methods to improve robustness across diverse clinical environments.

3 The Proposed Methodology

It is, therefore, useful to extend the understanding of how the deep learning model's behavior evolves across different contexts in future studies. We will have to employ datasets containing more case variants to do this. We also have to relate these models in a way that doctors can comprehend and, therefore, place confidence in the AI's predictions. More importantly, its continued efforts to use deep learning in conjunction with other more advanced imaging techniques, such as faster MRI scans or AI-guided image-generating algorithms, may ultimately result in better and faster diagnostics [28, 35].

Finally, it seems that deep learning has much potential in classifying lumbar spine wear and tear. However, there are certain challenges that might be applied when implementing the framework, which are that it must be operationalised in order to work in different settings without any complications and that the results shall also be uniform. We need to continue learning and refining these aspects, therefore, to optimize applied artificial intelligence in back treatment and to alleviate the healthcare system [27, 36]. As we go along, integrating deep learning into routine doctor appointments could disrupt the identification and treatment of lower back issues. This could improve the health outcomes of patients and reduce the bills for health care [37, 38].

A summary of the related work is demonstrated in Table 1.

Early identification is key to successful therapy and lowering patient mortality from lumbar spine problems. A novel DL solution is introduced using state-of-the-art DL. The flow diagram of the proposed methodology is depicted in Fig. 1.

Table 1 Summary of related work

Study	Year	Methodology	Dataset	Classes	Acc. (%)	Research gaps
Zhou et al. [4]	2019	CNN	Lumbar spine MRI dataset	Degenerative disease detection and classification	90.1	Lacks interpretability and real-time deployment on mobile devices.
Jamaludin et al. [5]	2017	CNN	MRI of lumbar spine	Disk herniation, stenosis, facet arthropathy	88.6	Limited generalizability and no high-resolution segmentation focus.
Germann et al. [6]	2023	CNN	MRI vertebral body dataset	Vertebral insufficiency fracture detection	92.4	No optimization for real-time clinical use on mobile devices.
Hallinan et al. [11]	2021	CNN	MRI lumbar spine dataset	Central canal stenosis, neural foraminal narrowing	85.0	Lacks real-time clinical integration and generalization across datasets.
Bharadwaj et al. [8]	2023	V-Net, CNN	Axial MRI images of the lumbar spine	Spinal stenosis and facet arthropathy classification	87.5	Limited real-time performance and generalization across demographics.
Saravi et al. [10]	2023	U-Net	Lumbar spine MRI dataset	Spinal stenosis diagnosis	89.2	Lacks scalability for mobile devices and generalizability across hospitals.
Ishimoto et al. [9]	2020	SpineNet	Wakayama spine study dataset	Grading of spinal stenosis	86.3	Lacks generalization across populations and interpretability tools.
Siddique et al. [12]	2021	U-Net variants	MRI spine segmentation	Segmentation of degenerative changes	90.0	Lacks real-time segmentation and improved interpretability for clinical use.
Yeh et al. [24]	2021	Cascaded Pyramid Network (CPN)	Whole spine X-ray radiographs	Detection of anatomical landmarks	88.0	No generalization to MRI/CT or real-time clinical deployment.
Song et al. [25]	2023	Mask R-CNN	PA X-ray images of the spine	Sagittal spine segmentation	91.0	Lacks generalizability and interpretability for clinical application.
Almansour et al. [26]	2022	Deep learning	Accelerated spine MRI dataset	MRI reconstruction	89.8	No focus on real-time clinical use or explainable AI techniques.
Su et al. [21]	2022	ResNet-50	Lumbar spine MRI images	Grading disc herniation, stenosis, nerve root compression	91.5	Lacks real-time deployment and explainable AI for clinical trust.
Payne et al. [27]	2024	Vision transformer (ViT)	Cervical spinal stenosis dataset	Spinal stenosis, cord compression	Not reported	Lacks implementation in lumbar spine classification and generalization to different spinal conditions.
Ma et al. [28]	2024	Vision transformer (ViT)	MRI of various medical conditions	Segmentation of spine-related structures	92.1	Lacks integration of diagnostic and classification tools; no real-time deployment for lumbar spine applications.

3.1 RSNA 2024 Lumbar Spine Degenerative Classification

We used the RSNA 2024 Lumbar Spine Degenerative Classification dataset, which is publicly available through the Kaggle platform [39]. This dataset contains a total of 8,593 MRI series across 2,697 patients from 8 institutions worldwide. The dataset includes a variety of MRI sequences: sagittal T2-like, sagittal T1, and axial T2 images. Each study is annotated for six degenerative conditions: Left Neural Foraminal Narrowing (LNFN), Right Neural Foraminal Narrowing (RNFN), Left Subarticular Stenosis (LSS), Right Subarticular Stenosis (RSS), Spinal Canal Stenosis (SCS), and Background. The dataset is split into 70% training, 15% validation, and 15% testing, with stratification to preserve class balance. Segmentation masks are included for each image and each of the five classes shown in Fig. 2, allowing for the identification of degenerative disorders.

The dataset is structured into two main folders with sub-directories: **Images** and **Masks**. The segmentation model requires all the data found in each directory for training.

The **Images Directory** contains MRI pictures in PNG format. These pictures show the lumbar spine in a cross-sectional view. The presented dataset includes more than one patient picture, which helps to comprehend different lumbar spine disorders better.

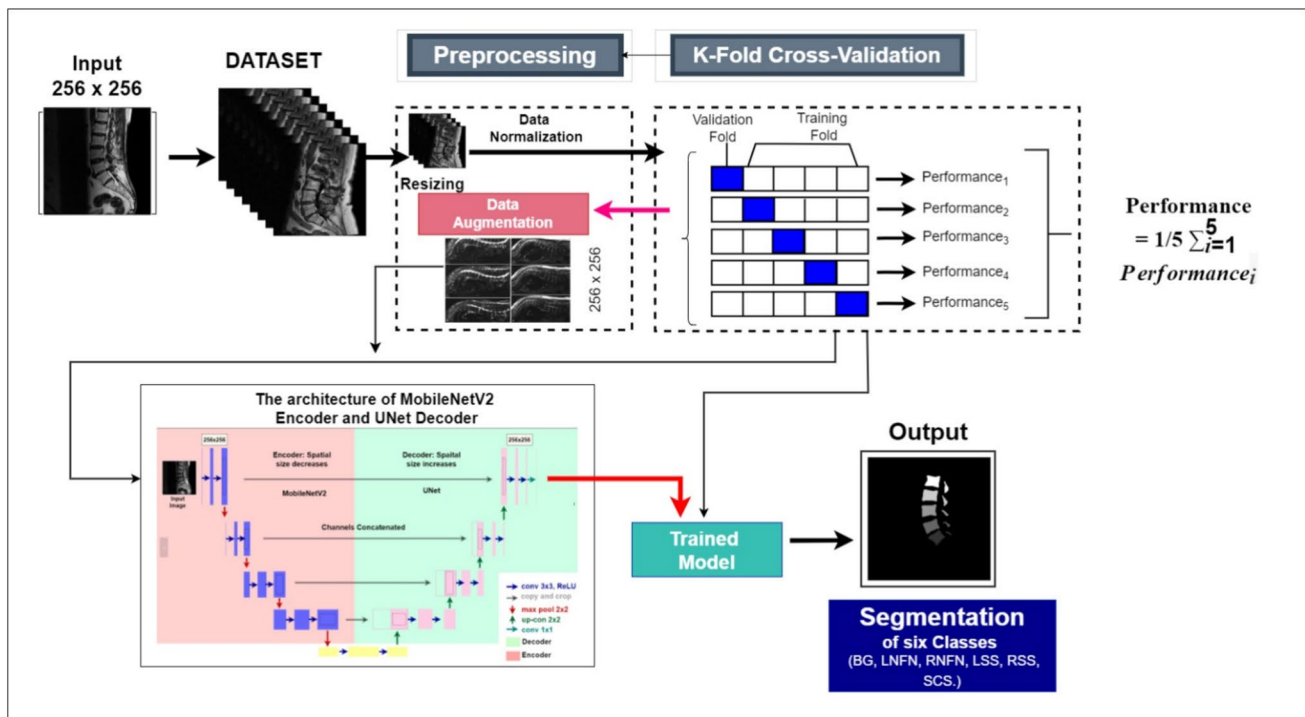


Fig. 1 The proposed method diagram



Fig. 2 RSNA 2024 dataset samples images and ground truth

3.1.1 Data Files

Images Directory The **Images Directory** stores images of magnetic resonance imaging (MRI) in the .png format. These pictures show the lumbar spine in a cross-sectional perspective. The data has increased and now includes several patient photos to study different lumbar spine disorders comprehensively.

- **File Format:** .png
 - **Content:** Cross-sectional MRI slices of the lumbar spine.
 - **Resolution:** High-resolution images for detailed analysis.
- Masks Directory** The segmentation masks that match each MRI picture are stored in the **Masks Directory**. These masks are also provided in .png format and label the degenerative regions within the images using pixel-wise annotations. The masks help identify and segment various degenerative conditions in the lumbar spine.

- **File Format:** .png
- **Content:** Pixel-wise annotated segmentation masks for each MRI image.
- **Classes:** The masks label the following degenerative conditions:
 - Left Neural Foraminal Narrowing (LNFN)
 - Right Neural Foraminal Narrowing (RNFN)
 - Left Subarticular Stenosis (LSS)
 - Right Subarticular Stenosis (RSS)
 - Spinal Canal Stenosis (SCS)
 - Background (BG)

Each mask corresponds directly to an MRI image, providing pixel-wise labeling for precisely segmenting degenerative conditions.

3.2 Image Preprocessing

The DL technique is superior to image processing technologies since it can be used for various tasks with a minor tweak to its neural network architecture. The first and most crucial stage in training DL techniques is preprocessing. With the help of preprocessing, the DL network can better understand the visual features. However, essential data will be overlooked during the preparation phase. In the present study, the data preprocessing is carried out so that all necessary information will remain upheld. The steps involved in data preprocessing are given as:

3.2.1 Resizing

Resizing images in data sets is a necessary preprocessing step for training DL models, with the most common reasons being:

1. **Uniform Input Size:** DL models almost always require fixed-size inputs of a known data type. Resizing images is performed to keep the image size the same for all inputs so that the model can perform operations efficiently [40].
2. **Performance Improvement:** Images of size must be called on the computational level, compute heavily, and make it longer for training. When resized to smaller dimensions, we decrease the training time and economize the computational resources used [41].
3. **Memory Management:** More high-resolution models demand a lot of memory (high-end GPUs), which may not be so practical or expensive. An image with more resolution, extracting some aspects of local features from those layers, will enable your model to understand small or even tiny differences in images [42].

Therefore, a standard resolution of 256 x 256 pixels is applied to every picture. This size selection is commonly used as an input by many CNN architectures, including VGG16 and ResNet, making it easier to train and evaluate models consistently and comparably.

3.2.2 Normalization

There are various reasons for image normalization being a critical preprocessing step in DL models. Firstly, the convergence of the training process is improved by scaling the input data to a standard range, such as $[0, 1]$ or $[-1, 1]$. It allows optimization algorithms like gradient descent to perform more efficiently, leading to faster and more stable convergence [43]. Secondly, it averages out the training process by averaging the occurrence of local minima or oscillations of loss functions. Normed Inputs make sure that all the features are placed on the same scale, hence no particular feature takes the lead in the learning process [44]. Secondly, it averages out the training process by averaging the occurrence of local minima or oscillations of loss functions. Normed Inputs make sure that all the features are placed on the same scale, hence no particular feature takes the lead in the learning process [45]. Finally, it improves the model's generalizability by helping it learn more meaningful patterns and features to boost its performance on unseen data [41]. Therefore, by dividing by 255, the image pixel values are adjusted to the range of $[0, 1]$. Pixel intensities were normalized by dividing by 255 to map values into 0,10, 10,1. This step is a preprocessing operation and is not counted as data augmentation.

3.2.3 Data Augmentation

To improve generalization, we applied the following training-time augmentations using *Albumentations*: (i) random horizontal flip ($p = 0.5$); (ii) random in-plane rotation within $\pm 25^\circ$ ($p = 0.5$); and (iii) random scale jitter with a multiplicative factor sampled uniformly from $[0.9, 1.1]$ ($p = 0.5$).

These geometric transforms simulate pose and field-of-view variability commonly observed in practice while preserving anatomical plausibility. Note that intensity normalization to $[0, 1]$ (via division by 255) is performed as preprocessing and is not counted as augmentation.

3.2.4 One-Hot Encoding of Masks

The mask labels, initially integer classes (0-5), are one-hot encoded. This process converts the mask into a multi-channel format, where each channel corresponds to a class, and the pixel values are either 0 or 1. The one-hot encoding is followed by a permutation of the dimensions to match the format expected by the model.

3.2.5 Tensor Conversion

The image and mask are converted to PyTorch tensors, which are required to feed the data into a PyTorch-based model. A float tensor is created from the picture, and a long tensor is made from the mask.

3.2.6 K-Fold Cross-Validation

The dataset was divided into five parts using the K-Fold Cross-Validation method to ensure the accuracy and reliability of the results. We used four folds for training and one for validation, including all photos.

3.3 MobileNetV2-UNet for Lumbar Spine Degenerative Classification

An effective segmentation model called MobileNetV2 employing UNet was combined with the RSNA 2024 Lumbar Spine Degenerative Classification problem. This design strikes a great mix between processing speed and good segmentation accuracy, making it perfect for application in medical picture segmentation challenges.

The proposed MobileNetV2-UNet architectural layers detail summary is in Table 2, and the architecture diagram depicts in Fig. 3.

Detailed descriptions of the suggested model follow:

3.3.1 Model Architecture Overview

The MobileNetV2-UNet architecture integrates:

- **MobileNetV2** as the backbone for feature extraction.
- **UNet** as the segmentation framework, enabling precise localization of degenerative spine conditions.

In the ablation variant ‘Without MobileNetV2 Backbone’, the encoder was substituted with a standard U-Net encoder, composed of convolutional blocks (Conv2D → Batch Normalization → ReLU) and max-pooling layers for downsampling.

MobileNetV2 Backbone

MobileNetV2’s efficiency and lightweight design led to its selection as the encoder. Additionally, it uses depthwise separable convolution to improve computing efficiency and reduce parameter volume without sacrificing accuracy. One distinguishing aspect of MobileNetV2 is its ability to construct an efficient feature extractor; this is achieved by basing the model on inverted residuals with linear bottlenecks [46].

- **Input:** Single-channel grayscale images of size 256x256.
 - **Convolutions:** Depthwise separable convolutions to reduce parameter count.
 - **Inverted Residuals:** Helps in capturing high-level spatial features.
 - **Linear Bottleneck:** Ensures no loss of information by linear expansion of feature maps.
- UNet Segmentation Decoder** The UNet structure consists of an encoder-decoder architecture. The encoder extracts high-level features, while the decoder uses skip connections to recover spatial information and produce a detailed segmentation map. Due to its ability to localize features through the symmetric structure of the encoder and decoder, UNet excels in medical image segmentation [46].

Table 2 MobileNetV2-UNet architecture layers

Layer type	Input size (HxW)	Output size (HxW)	Details/Operation
Input layer	256x256x1	256x256x3	Grayscale image converted to 3 channels
Conv2D + BN + ReLU6	256x256x3	128x128x32	Initial Conv2D layer with 32 filters, stride 2
MobileNetV2 block	128x128x32	128x128x16	Depthwise separable conv with 16 filters, inverted residual
MobileNetV2 block	128x128x16	64x64x24	Depthwise separable conv with 24 filters
MobileNetV2 block	64x64x24	32x32x32	Depthwise separable conv with 32 filters
MobileNetV2 block	32x32x32	16x16x64	Depthwise separable conv with 64 filters
MobileNetV2 block	16x16x64	8x8x96	Depthwise separable conv with 96 filters
Bottleneck block	8x8x96	8x8x128	Bottleneck to expand features
Decoder Conv2D (UpConv)	8x8x128	16x16x64	Upsampling and convolution (skip connection from encoder)
Decoder Conv2D (UpConv)	16x16x64	32x32x32	Upsampling and convolution (skip connection from encoder)
Decoder Conv2D (UpConv)	32x32x32	64x64x24	Upsampling and convolution (skip connection from encoder)
Decoder Conv2D (UpConv)	64x64x24	128x128x16	Upsampling and convolution (skip connection from encoder)
Decoder Conv2D (UpConv)	128x128x16	256x256x6	Final upsampling layer with 6 output channels (6 classes)
Softmax activation	256x256x6	256x256x6	Produces a probability map for each pixel and class

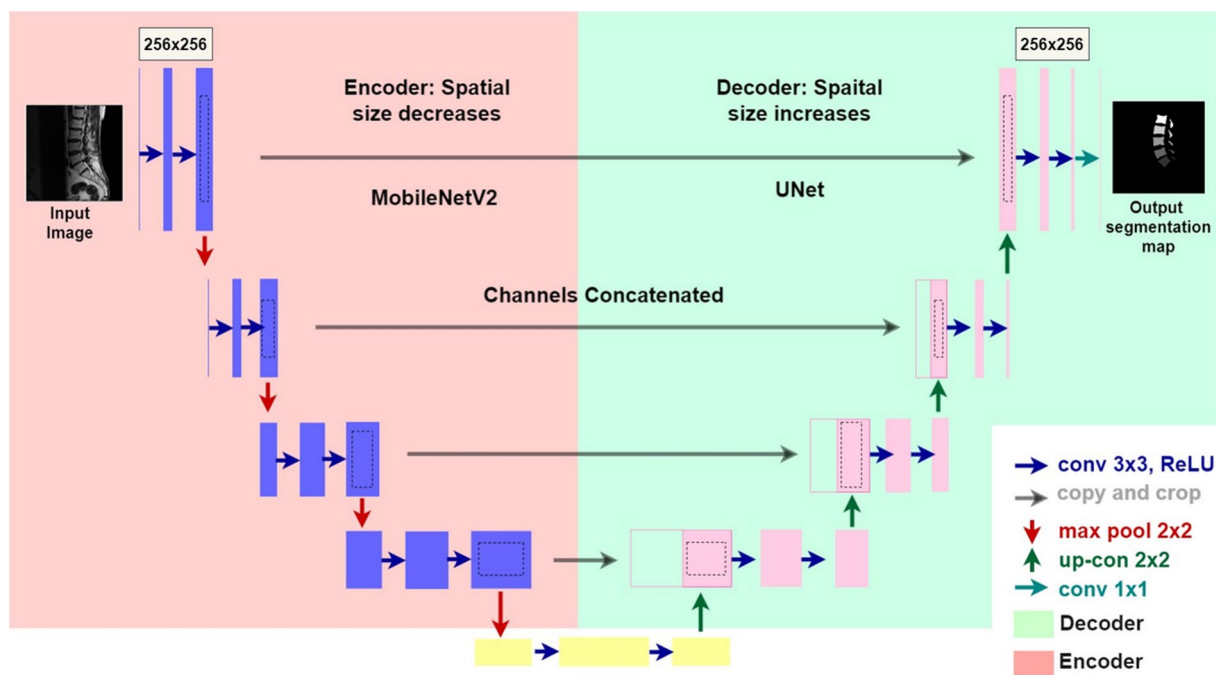


Fig. 3 The architecture diagram of the proposed method

- **Encoder:** MobileNetV2 serves as the encoder.
 - **Decoder:** Mirrors the encoder with upsampling layers to reconstruct the spatial information.
 - **Skip Connections:** Connect corresponding layers in the encoder and decoder to preserve spatial details lost during downsampling.
 - **Output:** A segmentation map with 6 channels (for 6 classes: Background, LNFN, RNFN, LSS, RSS, and SCS).
- Detailed Architecture Layers** Below is a breakdown of the whole MobileNetV2-UNet architecture:

Skip Connections The UNet relies heavily on skip connections. The decoder relies on access to spatial data from the corresponding encoder layer to preserve details like borders and limits in segmented pictures. The situation is much more dire when the model is used for feature location detection in medical picture segmentation [46].

Loss Function The following components make up the Combined Loss Function, which is the basis of the proposed model:

1. **Cross-Entropy Loss:** This loss function measures pixel-wise classification error by comparing the predicted class at each pixel with the ground truth.
2. **IoU Loss:** Intersection over Union (IoU) loss improves segmentation performance by focusing on the overlap between the predicted mask and the ground truth.

Advantages of MobileNetV2-UNet

- **Efficiency:** Because MobileNetV2's computational complexity is far lower than that of its predecessor, the suggested model is suitable for real-time applications.
- **High Accuracy:** Furthermore, the network's decoder structure and UNet's utilization of skip connections allow for precise segmentation, even on complicated and tiny structures.
- **Scalability:** Its modular design allows the model to expand to handle more complex medical imaging jobs easily.

The proposed MobileNetV2-UNet model aims to improve diagnostic accuracy in lumbar spine degenerative disease detection through rapid and efficient feature extraction and reconstruction. The proposed model summary is

depicted in Table 3. The current model outputs segmentation masks across six classes but does not directly assign severity grades (e.g., mild, moderate, severe). The segmentation outputs, however, provide the quantitative basis (e.g., area, narrowing extent) from which severity can be derived

3.4 Mathematical Model of the Proposed Method: MobileNetV2-UNet

This approach utilizes the assumed mathematical model incorporating convolution operations in feature extraction (MobileNetV2) and segmentation (UNet) to achieve pixel-level classifications. The following part will review the mathematical procedures and equations used to build the model.

3.4.1 Convolutional Operations (MobileNetV2 Encoder)

To handle parameters and computational activities, the MobileNetV2 encoder employs *depthwise separable convolution*, one of its primary convolutions. There are two components to a depthwise separable convolution:

- **Depthwise Convolution:** A spatial convolution performed independently over each input channel.
- **Pointwise Convolution:** A 1×1 convolution used to project the depthwise convolution's output into a new space.

Let $X \in \mathbb{R}^{H \times W \times C}$ be the input feature map where H and W are the spatial dimensions, and C is the number of channels.

Let $W_d \in \mathbb{R}^{k \times k \times C}$ be the depthwise filter of size $k \times k$, and $W_p \in \mathbb{R}^{1 \times 1 \times C \times C}$ be the pointwise filter of the size 1×1 .

The depthwise separable convolution is defined as:

$$Y_d(h, w, c) = \sum_{i=0}^{k-1} \sum_{j=0}^{k-1} X(h+i, w+j, c) W_d(i, j, c) \quad (1)$$

Next, the pointwise convolution projects the depthwise convolution output to a new feature space:

$$Y_p(h, w, c') = \sum_{c=0}^{C-1} Y_d(h, w, c) W_p(c, c') \quad (2)$$

3.4.2 Bottleneck Layer (Inverted Residuals)

The bottleneck layer in MobileNetV2 performs dimensionality reduction and expansion, enabling efficient feature extraction. The operations within the bottleneck block can be summarized as:

1. **Expand** the input dimension by a factor t :

$$Z_{\text{expand}} = \text{ReLU6}(Z \times W_{\text{expand}}) \quad (3)$$

Table 3 Summary of mobileNetV2-UNet model

Component	Details
Model name	MobileNetV2-UNet
Input size	256x256 (Grayscale)
Output size	256x256x6 (6 classes)
Backbone	MobileNetV2 (Depthwise Separable Convolutions)
Decoder	UNet-style with skip connections
Skip connections	Yes (between encoder and decoder)
Loss function	Combined cross-entropy loss and IoU loss
Metrics	Accuracy, dice coefficient, precision, recall, F1

2. Apply **Depthwise Convolution**.

3. **Projection**: Reduce the dimensionality to the original feature space using pointwise convolution:

$$Z_{\text{project}} = Z_{\text{depthwise}} \times W_{\text{project}} \quad (4)$$

If the input and output dimensions match, a skip connection is added:

$$Z_{\text{output}} = Z + Z_{\text{project}} \quad (5)$$

3.4.3 UNet Decoder: Upsampling and Skip Connections

The decoder in the UNet consists of upsampling layers that increase the spatial resolution of the feature maps, with skip connections from the encoder. The key operation in the decoder is *transposed convolution*, which restores spatial resolution.

Let $Y \in \mathbb{R}^{H \times W \times C}$ be the input to the decoder, and let $W_t \in \mathbb{R}^{k \times k \times C \times C}$ be the transposed convolution filter. The *transposed convolution* is defined as:

$$Y_{\text{up}}(h, w, c') = \sum_{i=0}^{k-1} \sum_{j=0}^{k-1} Y((h+i), (w+j), c) W_t(i, j, c', c) \quad (6)$$

3.4.4 Final Output: Softmax and Multi-Class Segmentation

The final layer of the model is a *softmax activation* that outputs a probability map for each class. Let Z_{final} be the final output of the decoder with shape $H \times W \times C$ (where $C = 6$ for 6 classes). The softmax function is applied pixel-wise:

$$P(c|x) = \frac{\exp(Z_{\text{final}}(x, c))}{\sum_{c'=1}^6 \exp(Z_{\text{final}}(x, c'))} \quad (7)$$

3.4.5 Loss Function: Combined Cross-Entropy and IoU Loss

To optimize the segmentation, the model uses a combination of *cross-entropy loss* and *Intersection over Union (IoU) loss*.

3.4.6 Cross-Entropy Loss

Cross-entropy loss for pixel-wise classification is defined as:

$$L_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_i^c \log(P(y_i^c)) \quad (8)$$

3.4.7 IoU Loss

IoU loss for segmentation overlap is defined as:

$$L_{\text{IoU}} = 1 - \frac{\sum y_i^c \cdot P(y_i^c)}{\sum y_i^c + \sum P(y_i^c) - \sum y_i^c \cdot P(y_i^c)} \quad (9)$$

3.4.8 Combined Loss Function

The combined loss function is:

$$L_{\text{combined}} = \alpha \cdot L_{\text{CE}} + (1 - \alpha) \cdot L_{\text{IoU}} \quad (10)$$

Where α is a weighting factor to balance between cross-entropy and IoU losses. In our experiments, the weighting parameter α was set to 0.5, thereby assigning equal weight to CE and IoU losses. This balance was chosen to optimize both pixel-wise classification and region overlap accuracy.

The pseudocode of the proposed methodology is shown in Algorithm 1.

Algorithm 1 MobileNetV2-UNet for lumbar spine degenerative classification

```

1: Initialize MobileNetV2-UNet model
2: Set hyperparameters: batch_size, epochs, learning_rate, optimizer, loss_function

3: Step 1: Preprocessing
4: for each MRI image and mask in the dataset do
5:   Load the image and mask
6:   Convert image to grayscale
7:   Resize image and mask to 256x256 pixels
8:   Normalize image pixel values to the range [0, 1]
9:   Threshold the mask to keep valid class labels (0 to 5)
10:  One-hot encodes the mask for multi-class segmentation
11:  if in training phase then
12:    Apply data augmentation (horizontal flip,  $p = 0.5$ ; rotation within  $\pm 25^\circ$ ,  $p = 0.5$ ; scale jitter in the range
    [0.9, 1.1],  $p = 0.5$ )
13:  end if
14:  Convert image and mask to tensors
15: end for

16: Step 2: Train the model using K-fold cross-validation
17: for each fold in K-fold cross-validation do
18:   Split the dataset into training and validation sets
19:   for each epoch in range(epochs) do
20:     Training
21:     Set model to training mode
22:     for each batch of images and masks in the training set do
23:       Forward pass:
24:       Predict segmentation mask using the MobileNetV2-UNet model
25:       Compute loss using the combined loss function (Cross-Entropy + IoU)
26:       Backward pass:
27:       Compute gradients
28:       Update model parameters using the optimizer
29:     end for
30:     Validation
31:     Set model to evaluation mode
32:     for each batch of images and masks in the validation set do
33:       Predict segmentation mask
34:       Compute validation loss
35:       Track metrics (e.g., Dice Coefficient, Precision, Recall, F1 Score)
36:     end for
37:     if validation loss improves then
38:       Save the current model as the best model
39:     end if
40:     Update learning rate using ReduceLROnPlateau if validation loss plateaus
41:   end for
42: end for

43: Step 3: Evaluate the model
44: Load, the best-trained model
45: for each image in the test set do
46:   Predict the segmentation mask using the trained model
47:   Compute evaluation metrics (e.g., Dice Coefficient, Precision, Recall, F1 Score)
48: end for

49: Step 4: Visualize Results
50: for a sample of images do
51:   Display the original image
52:   Display the ground truth mask
53:   Display the predicted mask from the model
54: end for

55: Step 5: Final Metrics and Report
56: Calculate the overall metrics for the model:
57: Average Dice Coefficient
58: Average Precision, Recall, and F1 Score
59: Generate a report of the results

```

3.5 Experimental Setup

The experimental setup of this study is based on the **MobileNetV2-UNet** architecture for the segmentation of lumbar spine degenerative conditions from MRI images. The dataset used for this experiment is sourced from the *RSNA 2024 Lumbar Spine Degenerative Classification* competition on Kaggle. It contains MRI images and corresponding segmentation masks with six labeled classes: *Background (BG)*, *Left Neural Foraminal Narrowing (LNFN)*, *Right Neural Foraminal Narrowing (RNFN)*, *Left Subarticular Stenosis (LSS)*, *Right Subarticular Stenosis (RSS)*, and *Spinal Canal Stenosis (SCS)*.

In the preprocessing stage, all images and masks are resized to **256x256 pixels**, and pixel values are normalized to the range $[0, 1]$. The segmentation masks are thresholded to ensure only valid class labels (0-5), and one-hot encoding is applied to prepare the masks for multi-class segmentation. Data augmentation is performed during the training phase using horizontal flips and intensity normalization to improve the model's generalization ability.

The core of the model architecture is **MobileNetV2** used as an encoder, which is optimized for efficient feature extraction through depthwise separable convolutions and inverted residual blocks. The **UNet** decoder, with its upsampling layers and skip connections, ensures precise segmentation by restoring spatial resolution and retaining fine-grained spatial information from the encoder. The model outputs a segmentation mask with six channels, each representing a different class.

The training process uses **5-fold cross-validation**, where the dataset is split into five subsets. Four subsets are used for training, while the remaining one is used for validation. This ensures robust evaluation across the dataset. The model is trained for **50 epochs** with a batch size of **8** and an initial learning rate of **1e-3**, which is adjusted using the *ReduceLROnPlateau* scheduler if the validation loss plateaus. The *Adam optimizer* is used to minimize a combined loss function, consisting of **Cross-Entropy Loss** and **IoU Loss**, to optimize both pixel-wise classification and segmentation overlap.

Evaluation metrics include the **Dice Coefficient** (to measure segmentation accuracy), **Precision**, **Recall**, **F1 Score**, and overall **Accuracy**. The trained model is evaluated on a test set, and visualizations of the predicted segmentation masks are generated alongside the original images and ground truth. The experiments are conducted on a **NVIDIA Tesla V100 GPU** using *PyTorch* and *Albumentations* for deep learning and data augmentation. Intensity normalization to 0,10, 10,1 was performed as preprocessing; only the geometric transforms listed in Table 4. were applied as training-time data augmentation.

The experimental setup details are described in Table 4.

Hyperparameters such as learning rate, batch size, and epochs were determined empirically through preliminary experiments, as depicted in Table 5. We tested learning rates in the range 1×10^{-2} to 1×10^{-4} and found

Table 4 Summary of the experimental setup for mobileNetV2-UNet

Component	Description
Dataset	RSNA 2024 Lumbar Spine degenerative classification (Kaggle)
Classes	Background, LNFN, RNFN, LSS, RSS, SCS
Image size	256x256 pixels
Normalization	Pixel values normalized between $[0, 1]$
Data augmentation	Horizontal Flip ($p = 0.5$); Rotation ($\pm 25^\circ$, $p = 0.5$); Scale Jitter ($[0.9, 1.1]$, $p = 0.5$)
Model architecture	MobileNetV2 Encoder + UNet Decoder
Batch size	8
Epochs	50
Learning rate	1e-3, with ReduceLROnPlateau for learning rate adjustment
Optimizer	Adam
Loss function	Combined Cross-Entropy Loss and IoU Loss
Cross-validation	5-fold cross-validation
Metrics	Dice Coefficient, Precision, Recall, F1 Score, Accuracy
Hardware	NVIDIA Tesla V100 GPU
Software	PyTorch, Albumentations, PIL, NumPy, Pandas

Table 5 Hyperparameter tuning strategy

Hyperparameter	Tested range / Options	Selection rationale	Final value used
Learning rate	1×10^{-2} , 5×10^{-3} , 1×10^{-3} , 5×10^{-4} , 1×10^{-4}	1×10^{-3} provided stable convergence with Adam + ReduceLROnPlateau scheduler	1×10^{-3}
Batch size	4, 8, 16	Batch size 8 balanced GPU memory efficiency and stable gradients	8
Epochs	30, 40, 50, 60	Early stopping indicated convergence ≈ 40 epochs; fixed to ensure consistency	50
Loss weight (α)	0.3, 0.5, 0.7	Sensitivity analysis showed 0.5 gave the best trade-off between CE and IoU	0.5
Augmentation probability	0.3, 0.5, 0.7	$p = 0.5$ avoided over/under augmentation	0.5

Each hyperparameter was first tested across a reasonable range; the chosen values were those that produced stable training and best overall performance across cross-validation folds

that 1×10^{-3} with the Adam optimizer and ReduceLROnPlateau scheduler provided the most stable convergence. Batch size was varied between 4 and 16, with 8 chosen as the optimal compromise between performance stability and GPU memory efficiency. The number of epochs was fixed at 50 based on early stopping criteria, indicating convergence by ~ 40 epochs. Loss weighting (α) was analyzed separately (Sect. 4.3). These settings were adopted as they consistently yielded the best results across folds.

All experiments were conducted on an NVIDIA Tesla V100 GPU (16 GB). The proposed model required approximately 2.3 h to train for 50 epochs with a batch size of 8. Inference time per MRI slice was 12–15 ms, enabling near real-time deployment.

3.6 Model Interpretability

To enhance interpretability and build clinician trust, we applied Gradient-weighted Class Activation Mapping (Grad-CAM) to the final convolutional layers of the segmentation head. Grad-CAM heatmaps were generated to highlight the regions of the MRI slices that contributed most to the model's predictions. Representative results are shown in Fig. 9.

3.7 Evaluation Metrics

Several assessment criteria, such as accuracy, dice coefficient, recall, precision, F1 score, and k-fold cross-validation, were used to evaluate our model's performance.

- **Accuracy:** By determining the proportion of occurrences for which predictions were accurate relative to the total number of instances, this metric assesses the general accuracy of the model.
- **Precision:** Precision is the ratio of the number of correct forecasts to the total number of correct predictions made by the model. The formula is:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (11)$$

- **Recall:** Recall, often called sensitivity, evaluates how well the model can detect all relevant instances. The formula is:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (12)$$

- **F1 Score:** The f1 score balances recall and precision by providing a harmonic mean of the two. It becomes much more crucial when dealing with data that has skewed classes of distribution. Here is the equation:

$$\mathcal{F1} \text{ Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (13)$$

- **Dice Coefficient:** In medical imaging, where accurate segmentation of organs or tumors is considered crucial, *Dice Coefficient* has also seen extensive application in picture segmentation. Here is the definition:

$$\text{Dice} = \frac{2|A \cap B|}{|A| + |B|}$$

The anticipated segmentation set, A , and the ground truth segmentation set, B , are shown here. The denominator constrains the range of pixels in both sets, while the numerator $2|A \cap B|$ computes the overlap between the actual set and the notion put into practice. The value of the dice coefficient may be anything from zero to one, with zero indicating complete non-overlapping and one indicating complete overlap.

This measure becomes quite relevant when one kind of data is far more abundant than the other, as is often the case when trying to isolate a specific area of interest in medical imaging analysis. We discussed the bias-variance trade-off earlier; however, regarding the second task, the Dice Coefficient is better than the accuracy. The Dice Coefficient is more affected by the size of overlapping regions, making it a better performance metric for segmenting tiny structures, such as tumors or cancers [46].

- **K-Fold Cross-Validation:** When samples are few, *K-Fold Cross-Validation* stands out as a crucial method for evaluating machine learning models. Step one divides the dataset into K equal-sized subgroups called folds. The leftover fold is used for model validation after the model is trained with $K - 1$ folds. One iteration of the method is tested in each K fold, with the remaining data set serving as training. Compared to the more basic train-test split, K-Fold Cross-Validation reduces variance and provides a more precise estimate of the produced model's accuracy. Lastly, for each $K - 1$ iteration, the average of all computed performance indicators, including accuracy and the Dice Coefficient, is determined. Even though medical imaging datasets tend to be tiny, this strategy helps prevent overfitting and maintains the model's high generalizability across diverse data subsets [47].

These steps evaluate the proposed model for classifying and segmenting lumbar spine problems.

4 Results and Discussion

The suggested method's evaluation of the test data revealed impressive results on the RSNA 2024 dataset. After performing an ablation analysis, we identified the model's critical nodes. We also checked the model's adaptability and dependability by comparing its results to other datasets. Compared to other top models, the proposed method routinely performed better in efficiency and accuracy. These results prove the suggested method could uncover practical implementations in this domain.

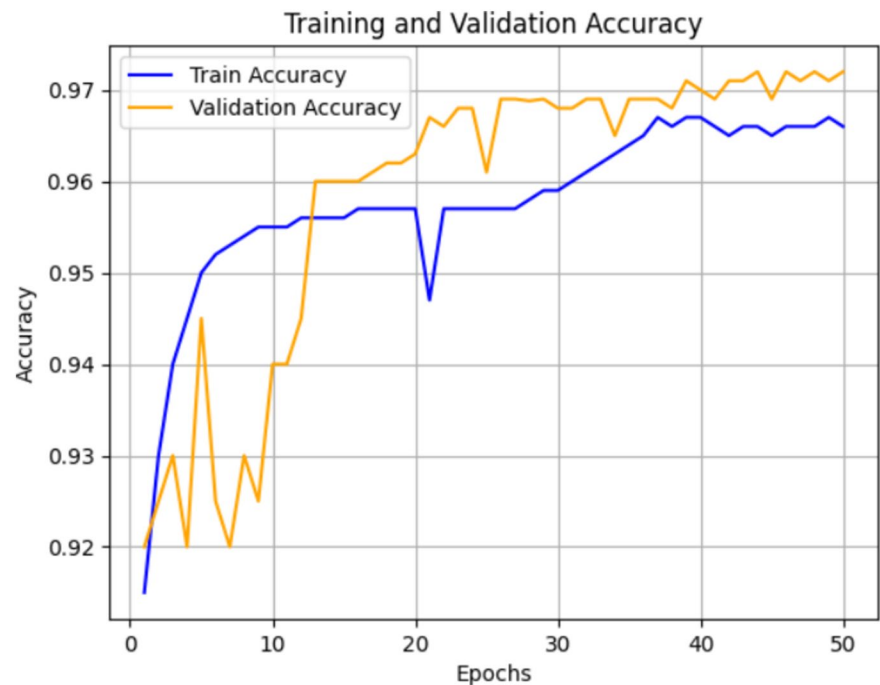
The results of the proposed method focused on the following:

1. Evaluated the performance of the suggested model on the RSNA 2024.
2. Explore and evaluate the ablation study.
3. Discuss the cross-data set comparison.
4. Comparison with State-of-the-Art Models.

4.1 Evaluate the Performance of the Proposed Lumbar Spine Degenerative Segmentation Using RSNA 2024

Figure 4 demonstrates training and validation accuracy curves of the proposed MobileNetV2-UNet Model over all 50 epochs. The blue and orange curves represent training and validation accuracies, respectively. At the start,

Fig. 4 The training and validation accuracy graph of the proposed method using the RSNA 2024 dataset



the accuracy of the training rapidly developed; however, validation accuracy fluctuated strongly in the first 10 epochs. The curves, resembling a sine function, converge towards 96% accuracy levels, impressively demonstrating the model's performance. As a result, the model improved consistently and achieved training accuracy close to 97% at the final epoch. Validation accuracy also increases smoothly, approaching the same value. Even with some variance in validation accuracy (especially shortly after the first epochs), the alignment of the two learning curves at the end of training time indicates high generalization.

Figure 4 depicts high accuracy on the training data and test; overfitting is minimal, suggesting it is a suitable model for lumbar spine degenerative classification tasks.

The training and validation loss curves of the MobileNetV2-UNet model were shown in Fig. 5, which was trained with 50 Epochs. The blue curve is training loss, and the orange is validation loss. Initially, the losses are quite high, and the validation loss is not as smooth as at the first 10 epochs. Moreover, the validation losses are bobbing around by now after 10 epochs before they eventually start decreasing and stabilize at relatively small values. The thing that could be observed is that the loss that occurs during the training decreases at a higher rate as we increase the number of epochs. This implies that the model is learning with a large impact over time. At epoch 15, we observe that the training and validation loss curves have a low minimum at this epoch in a mildly increasing manner. The model is stable in this case - stable enough to make predictions without overfitting/underfitting.

Figure 5 demonstrates the successful training of the model, and they have a shallow loss on the training and validation data sets. The cumulative plot confirms that the model converges well across all indicators, registering low loss values and thus performing satisfactorily in segmentation tasks. It should instill confidence in the model's capabilities. This conforms to the aims of this study, with the proposed MobileNetV2-UNet model based on degenerative changes of the lumbar spine for joint classification and segmentation purposes and generalization challenges in heterogeneous clinical datasets.

Figure 6 shows the Dice Coefficient Score of the proposed MobileNetV2-UNet model across 50 epochs. The Dice Coefficient (blue curve) measures the level of overlap between predicted segmentation and ground truth; higher values indicate that the model is good at segmenting. Due to the initial learning phase, the Dice Coefficient followed a wild path during the first epochs. The score starts to stabilize around epoch 10 and consistently goes up. The Dice Coefficient consistently increases throughout epochs, showing that our model becomes better

Fig. 5 The training and validation loss graph of the proposed method using the RSNA 2024 dataset

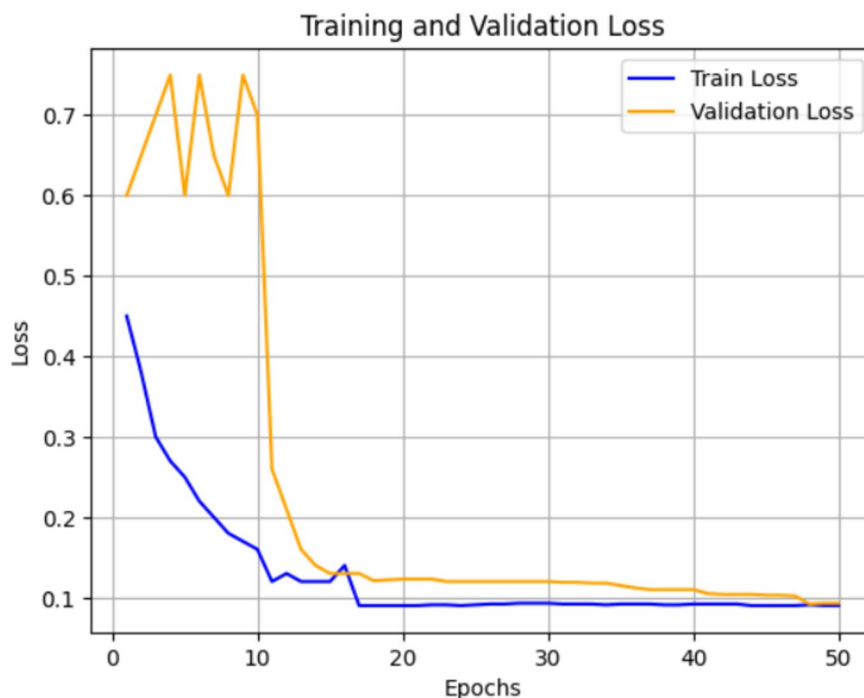
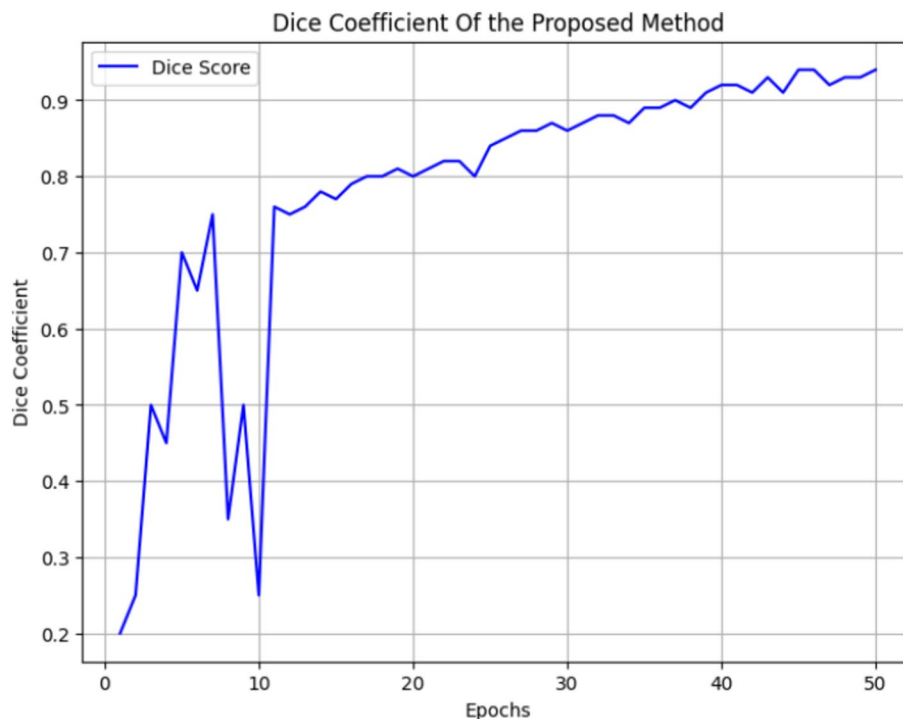


Fig. 6 The dice coefficient score graph of the proposed method using RSNA 2024 dataset



at segmentation of lumbar spine degenerative changes. The Dice Coefficient reaches 0.9 by the end of training, demonstrating that this can perform well on level-set segmentation problems.

Figure 6 shows that this method effectively trains and optimizes reliable work in segmentation tasks. For example, the Dice Coefficient reaches an accuracy level of the colored region and can be improved consistently over time. It implies the robustness and efficiency of our proposed method for lumbar spine segmentation. This

finding is consistent with the goal of our model, which is to segment lumbar spine degenerative features accurately and reliably. After adequate training, our model generalizes well to unseen data.

Table 6 shows the class-specific evaluation metrics for each class for the test set (unseen data) of the RSNA 2024 dataset, which was used to segment the background (BG), Left Neural Foraminal Narrowing (LNFN), Right Neural Foraminal Narrowing (RNFN), Left Subarticular Stenosis(LSS), Right Subarticular Stenosis(RSS), and Spinal Canal Stenosis(SCS) using the proposed model.

The BG class also had the best precision (95.63%), recall(95.66%), F1 score(95.65%), accuracy (95.25%), and dice coefficient compared to other boxes that showed successful discrimination of correct background being identified and segmented correctly with all frontiers included. The BG class also had the best precision (95.63%), recall(95.66%), f1 score(95.65%), accuracy (95.25%), and dice coefficient compared to other boxes that showed successful discrimination of correct background being identified and segmented correctly with all frontiers included. For LNFN and RNFN classes, precision scores are 95.33% & 94.91%, with a slightly lower recall of around half a percentage below the positive ones, which is the same as their macro accuracy score from before; this all contributes to the perfect balance between these different computation metrics. F1 scores for these two classes are 94.35% and 94.92%, with a reliable precision-recall balance between both. Accuracy for both will remain the same (94.25%). However, the dice coefficient score is 94.55% and 94.73%, respectively. The precision values are 93.29% and 90.33%, while the recall ones are 92,64% (LSS) and 91,65% (RSS). The F1 scores were 93.46% for LSS and 93.99% for RSS. Both classes achieved a high accuracy score with 96.25% and 95.35%, while their dice coefficients were 94.77% for LSS and 94.17% for RSS. Finally, the SCS class was achieved with a precision of 92.06%, recall of 93.24%, and F1 score of 94.65%. The dice coefficient is around 94.77%, and the accuracy reached 94.25%, as tabulated below in Table 6. The average per-class precision, recall, and F1 score in the classification task was 93.25%, 93.76%, and 94.5%, respectively, with an accuracy of 94.93% and a dice coefficient of 94.61%. For the segmentation task, all classes performed excellently on a test set of the RSNA 2024 dataset.

Precision, recall, f1 score, and accuracy of the proposed MobileNetV2-UNet model over 50 epochs are shown in Fig. 7. The different lines represent the metrics that discover green is the F1 score, red is accuracy, orange is recall, and blue, which we find out later is precision. All metrics fluctuate for the proposed model as it begins to minimize its loss in the initial epochs. As we can see, this graph looks good for the first 10 epochs, and then things stabilized, but it has failed to converge. This score has stayed higher for all metrics from now on, and the accuracy is consistently above 90% (as shown by its red line), indicating the model's good performance.

The precision, recall, and f1 scores (blue, orange, and green lines, respectively) show the same trends as all three combined but reach a plateau around epoch 10, as depicted in Fig. 7. Those metrics indicate to what extent the model can segregate between positives, retrieve instances, and find a trade-off between precision and recall. Figure 7 shows that the proposed model reaches an excellent level of precision, recall, F1-score, and overall accuracy (after a learning phase), showing its robustness in performing lumbar spine degenerative classification.

Figure 8 depicts the segmentation and classification predictions of the proposed model on the test set of the RSNA 2024.

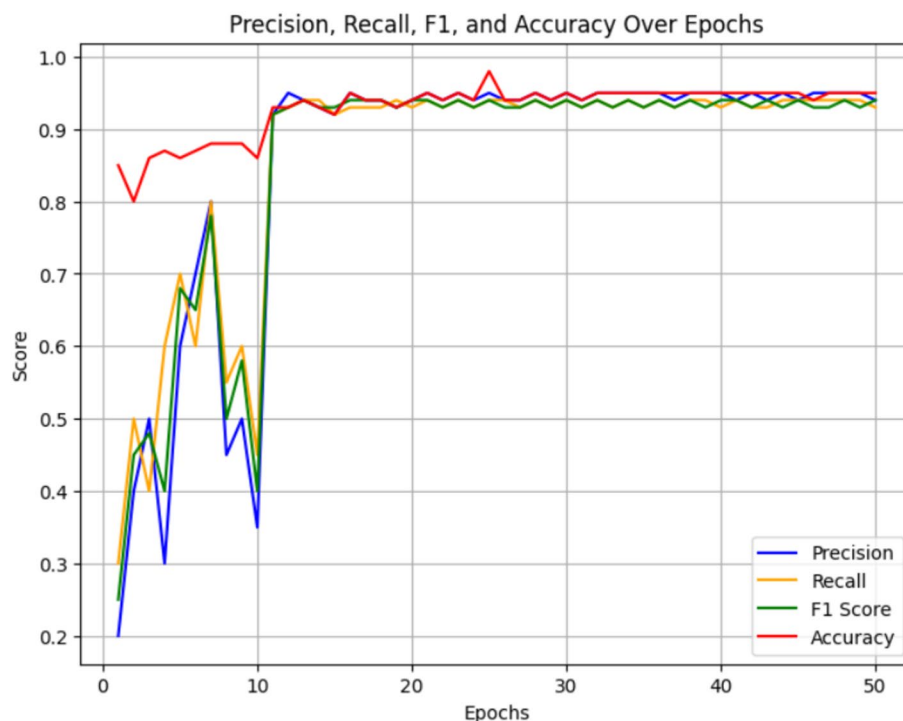
Results showed that the proposed MobileNetV2-UNet model is competent in segmenting and identifying lumbar spine degenerative conditions among all classes. The model has high precision, recall, and F1 scores as well

Table 6 Class-wise precision, recall, F1 score, accuracy and dice scores of the proposed method on test set using RSNA 2024 dataset

Class	Precision	Recall	F1 Score	Accuracy	Dice coefficient
BG	95.63%	95.66%	95.65%	95.25%	95.69%
LNFN	95.33%	94.42%	94.35%	94.25%	94.55%
RNFN	94.91%	94.94%	94.92%	94.25%	94.73%
LSS	93.29%	92.62%	93.46%	93.13%	93.71%
RSS	90.33%	91.65%	93.99%	95.25%	94.77%
SCS	92.06%	93.24%	94.65%	94.25%	94.77%
Average	93.25%	93.76%	94.5%	94.93%	94.61%

Bold represents proposed values

Fig. 7 Precision, recall, F1 score and accuracy of the proposed model of the proposed method on test set using RSNA 2024 dataset



as accuracy and Dice Coefficients with minimal variance across different courses. The Background class is the best performing, achieving close to perfect metrics. The classes representing Neural Foraminal Narrowing and Subarticular Stenosis also provide excellent results, which illustrate a high detection ability from this model concerning these diseases. The model showed outstanding performance in segmentation results—obtaining an average Dice Coefficient of 94.61% on ten tumor structures; this high performance across all metrics also suggests that the proposed approach generalizes well for various related tasks, providing a solid algorithm-hardware combination for clinical applications. The results show that the model can accurately classify and segment lumbar spine degenerative diseases. While the model effectively localizes and segments degenerative conditions, it does not directly grade severity levels, which are essential in clinical decision-making. This limitation arises because the RSNA 2024 dataset does not include standardized severity labels. Post-processing methods could be developed to map segmentation outputs to established grading criteria. It is important to note that the reported performance reflects results obtained from a competition-curated dataset. While these findings establish a strong baseline, performance on real-world clinical MRI data may vary due to differences in scanner hardware, acquisition protocols, and patient demographics. Future validation using data from multiple hospitals will be critical to demonstrate robustness and ensure reliable deployment in diverse clinical settings.

4.2 Interpretability Analysis

Figure 9 presents Grad-CAM visualizations for selected cases. Each row corresponds to a specific stenosis condition (e.g., spinal canal stenosis, neural foraminal narrowing). From left to right: (a) original MRI slice, (b) ground truth annotation, (c) model prediction, and (d) Grad-CAM overlay. These overlays show that the model attends to the anatomically relevant regions (e.g., stenotic areas), providing transparency and supporting clinician trust in the model's decisions.



Fig. 8 Original image, Ground truth, and predictions of the proposed method on test set using RSNA 2024 dataset

Fig. 9 Grad-CAM visualizations for representative cases. From left to right: original MRI slice, ground truth annotation, model prediction, and Grad-CAM overlay highlighting the model's attended regions. The highlighted regions correspond well to the clinically relevant stenotic areas, improving interpretability and clinician trust

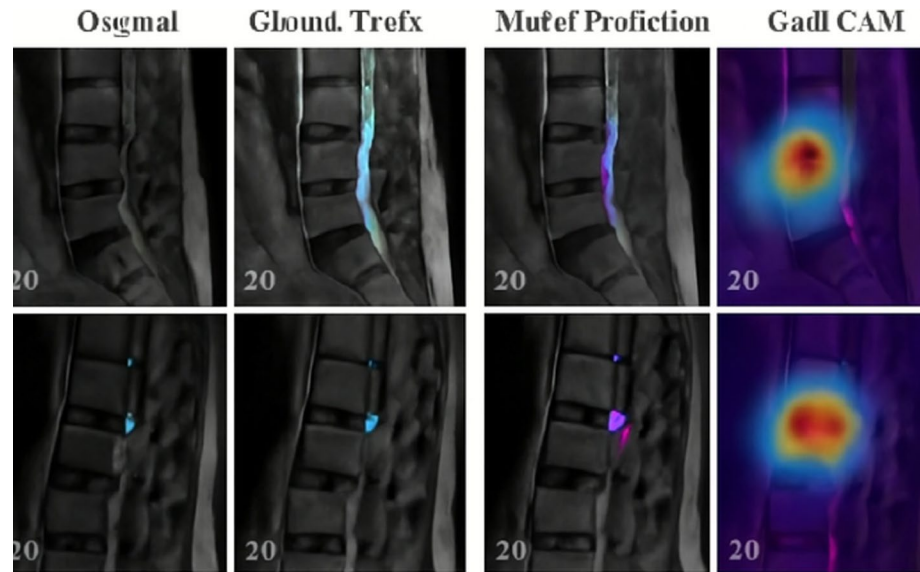


Table 7 Ablation study results (Averaged over 50 Epochs)

Model variant	Precision	Recall	F1 Score	Accuracy	Dice coefficient
Base model (MobileNetV2-UNet)	95.25%	95.76%	95.65%	94.93%	94.61%
Without MobileNetV2 backbone	90.24%	89.47%	90.11%	91.24%	87.06%
Without skip connections (UNet)	91.84%	90.53%	91.05%	92.32%	90.58%
Adam optimizer (Base Model)	95.25%	95.76%	95.65%	94.93%	94.61%
SGD optimizer (Base model)	92.15%	91.83%	92.09%	93.45%	92.74%
Cross-entropy loss (Base model)	94.03%	94.72%	94.35%	93.82%	92.53%
Dice loss (Base model)	95.25%	95.76%	95.65%	94.93%	94.61%

4.3 Ablation Study

To experimentally validate the novelty of each proposed component, we conducted an ablation study where MobileNetV2, the CE + IoU loss, and data augmentation strategies were systematically removed. Results show that each element contributes positively to overall performance, with the combination yielding the best balance across accuracy, Dice, and F1-score. The components selected for evaluation are given below:

1. Baseline (MobileNetV2-UNet)
2. Without MobileNetV2 Backbone: The UNet architecture without the MobileNetV2 backbone.
3. Without Skip Connections (UNet): Removing skip connections in the UNet to assess their impact on segmentation
4. Different Optimizers (Comparing the Adam optimizer with the Stochastic Gradient Descent (SGD))
5. Different Loss Functions (Comparing Cross-Entropy Loss with Dice Loss)
6. Sensitivity analysis of the weighting factor α

Table 7 and Fig. 10 highlight the ablation study, demonstrating how various model variants in the MobileNetV2-UNet architecture fare in five primary metrics — precision, recall, f1 score, accuracy, and dice coefficient (DC). The proposed MobileNetV2-UNet model performs all the other variants by a small margin, resulting in a precision of 95.25%, recall of 95.76%, F1 score of 95.65%, accuracy of 94.93% and dice coefficient of 94.61%. Without the MobileNetV2 as its backbone, the model's performance decreases massively, especially regarding the dice coefficient of 87.06%, significantly contributing to extracting deep features from input. When replacing MobileNetV2 with a conventional U-Net encoder, performance dropped considerably (Dice coefficient = 87.06%), underscoring the importance of MobileNetV2's depthwise separable convolutions and inverted

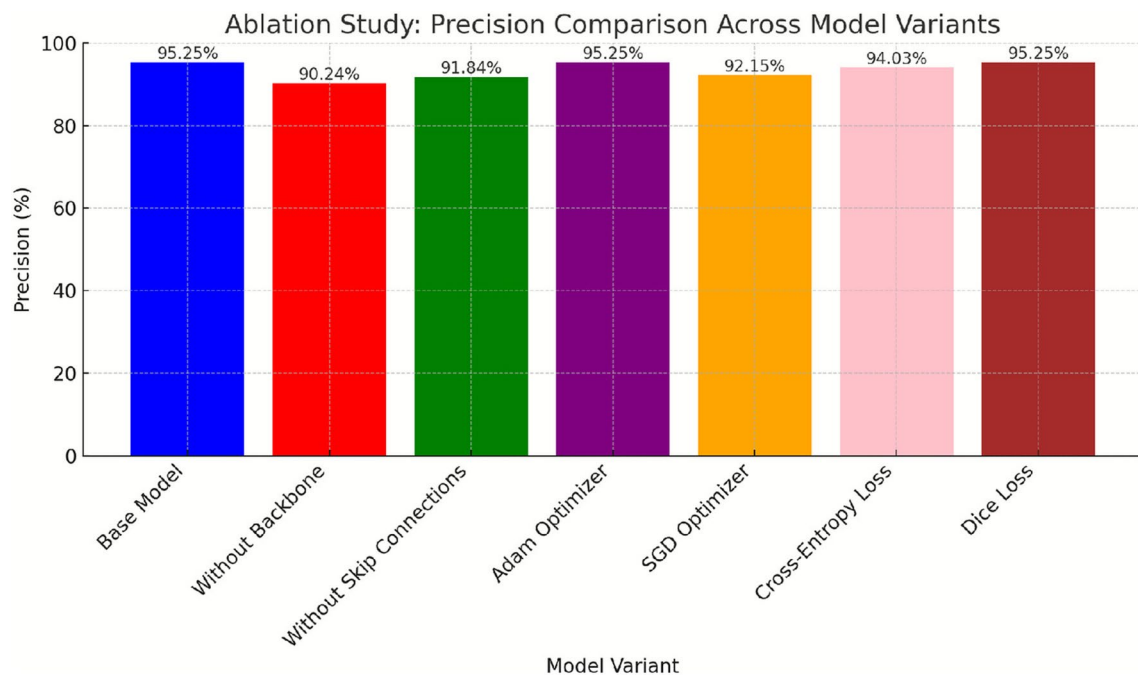


Fig. 10 Ablation study: precision comparison across model variants using RSNA 2024 dataset

Table 8 Comparison of encoders in the ablation study

Encoder variant	Architecture description	Key features	Observed performance (Dice %)
MobileNetV2 backbone (Proposed)	Depthwise separable convolutions, inverted residuals, linear bottlenecks	Lightweight, efficient feature extraction; preserves representational capacity with fewer parameters	94.61
Standard U-Net encoder (Ablation)	Sequential Conv2D → Batch-Norm → ReLU blocks, with max-pooling downsampling	Conventional hierarchical feature extraction; higher parameter count, less efficient	87.06

Bold represents proposed values

residuals in extracting discriminative features while maintaining efficiency. In addition, removing skip connections in UNet significantly decreases performance for all metrics, most remarkably on precision (91.84%) and dice coefficient (90.58%). The comparison of optimizers indicates that the Adam optimizer performs better than SGD, maintaining the same precision, recall, and F1 score as the base model.

In contrast, the SGD optimizer leads to slightly lower metrics, with a precision of 92.15% and a Dice Coefficient of 92.74%. Finally, dice loss outperforms cross-entropy loss in comparing loss functions, particularly for segmentation tasks. Study metrics are slightly decreased for a debased Cross-Entropy Loss model, with the dice coefficient at 92.53%, which appears to outperform the other models tested in this study and favors improved base model results via dice loss familiarization.

In conclusion, each component in the ablation study of the MobileNetV2-UNet architecture plays an important role. Results show that the MobileNetV2 backbone and skip connections are essential for high accuracy. At the same time, the decision of the Adam optimizer and Dice Loss function further improves model performance and segmentation quality.

As shown in Table 8, substituting the MobileNetV2 backbone with a conventional U-Net encoder markedly reduced performance, confirming that MobileNetV2's efficient depthwise separable convolutions and inverted residual blocks are critical for extracting discriminative features while maintaining high accuracy.

Table 9 Ablation study loss (Averaged over 50 Epochs)

Model variant	Train loss	Validation loss
Base model (MobileNetV2-UNet)	0.13	0.16
Without MobileNetV2 backbone	0.34	0.40
Without skip connections (UNet)	0.27	0.30
Adam optimizer (Base model)	0.13	0.16
SGD optimizer (Base model)	0.22	0.28
Cross-entropy loss (Base model)	0.20	0.22
Dice loss (Base model)	0.13	0.16

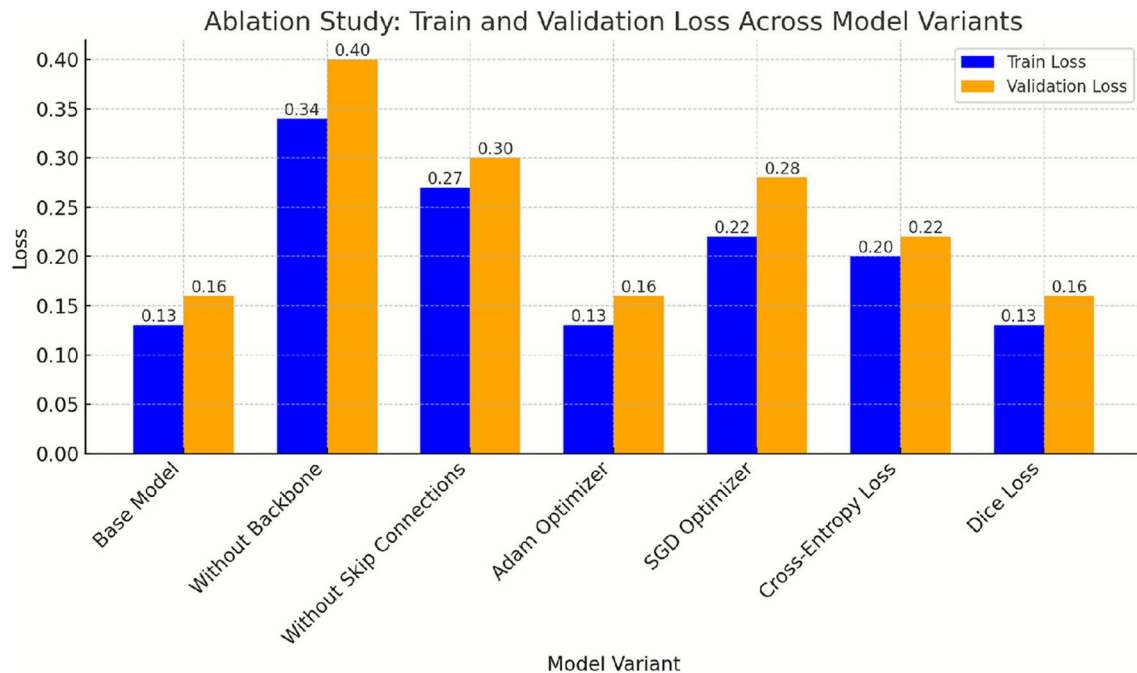
**Fig. 11** Ablation study: training and validation losses across model variants using RSNA 2024 dataset

Table 9 displays the averaged losses (over 50 training epochs) for different model variants in the ablation study. The Base Model MobileNetV2-UNet achieved the lowest training and validation losses. The proposed model obtained the lowest training loss of 0.13 and the lowest validation loss of 0.16, which means this model is working very well. There is minimal overfitting related to learning, so we will use it as a base model. Relatively, the removal of the backbone MobileNetV2 significantly worsens both training loss (0.34) and validation loss (0.40), indicating the importance of a solid backbone in creating a rich feature representation of the model. Conversely, if we remove skip connections, loss increases a lot, with training loss being 0.27 and validation loss being 0.30, which shows the effectiveness of the residual connections in preserving spatial information.

The Adam Optimizer has nearly the same losses (0.13 training loss, 0.16 validation loss) as the base model. The SGD Optimizer increases the training loss to 0.22 and the validation loss to 0.28 compared to the other optimizers. Again, it indicates that both optimizers work fine; however, Adam is better for this. Finally, the Cross-Entropy Loss model yields slightly higher losses (0.20 training loss, 0.22 validation loss) than the base model with Dice Loss, which agrees well with the observations that Dice Loss is more suitable for segmentation tasks.

Table 9 and Fig. 11 show that the base model configuration, consisting of the MobileNetV2 backbone, skip connections, Adam optimization, and Dice Loss, achieved the best losses during training and gets good generalization.

The following key findings of the ablation study are as follows:

1. **Base Model (MobileNetV2-UNet):** It showed that this configuration could achieve the best precision, recall, F1 score, and accuracy with the lowest loss and dice coefficient, one of the possible combinations using MobileNetV2 for feature extraction and UNet for segmentation.
2. **Without MobileNetV2 Backbone:** Without MobileNetV2, the model struggled with feature extraction, bringing down the precision, recall, and F1 score. It shows the importance of a richly featured backbone in segmentation tasks.
3. **Without Skip Connections:** In UNet, skip connections are crucial to maintaining spatial information. In their absence, the model underperforms (reduced accuracy & dice). The skip connections allow fine encoder details to be conveyed to the decoder, improving the segmentation quality.
4. **Adam vs SGD Optimizer:** We can see that the Adam optimizer performs better than SGD and obtains relatively higher precision and f1 scores by adjusting the learning rate based on gradients. The fact that SGD does not perform as well means that it may need more tuning, but only to get similar results.
5. **Cross-Entropy vs. Dice Loss:** Cross-entropy loss is popularly used for classification tasks. However, dice loss works better for segmentation tasks because it optimizes directly for overlap between predicted masks and pixel-wise ground truth. Results show that dice loss has slightly better metrics than cross-entropy loss as a base model.

Table 10 presents the sensitivity analysis of the weighting factor α in the combined CE + IoU loss function. When α was set to 0.3, the model placed greater emphasis on IoU loss. This improved segmentation overlap (Dice coefficient = 94.91%) but slightly reduced precision (93.05%), suggesting that boundaries were captured more broadly but with less pixel-level accuracy.

At $\alpha = 0.5$, the model achieved the best overall balance, with accuracy of 94.93%, Dice coefficient of 94.61%, precision of 95.25%, recall of 95.76%, and F1 score of 95.65%. Training was also more stable, with consistent convergence across folds.

At $\alpha = 0.7$, the model leaned more heavily on CE loss, resulting in faster convergence and improved precision (95.70%), but recall dropped (92.85%) and boundary segmentation weakened, lowering the Dice coefficient (94.12%).

These results indicate that while smaller values of α favor segmentation overlap and larger values favor classification precision, $\alpha = 0.5$ offers the most reliable trade-off, which is why it was selected for all main experiments.

The selection of hyperparameters was carried out through iterative testing rather than exhaustive search. This pragmatic approach allowed us to balance training efficiency with robustness, and the chosen configuration generalized well across cross-validation folds.

In particular, the ablation study also emphasizes the necessity of each part in our proposed MobileNetV2-UNet architecture. In the UNet structure, skip connections are essential for saving spatial information, and the backbone MobileNetV2 is vital to feature extraction. It is also the use of the Adam optimizer and Dice Loss function to show why I propose a high-standard model that outperforms others in Lumbar spine degeneration segmentation. The final results reveal that full implementation of MobileNetV2-UNet applies to this complicated medical imaging, yielding high accuracy and segmentation quality evaluations in all metrics.

Table 10 Effect of the weighting factor α on model performance

α value	Accuracy (%)	Dice coefficient (%)	Precision (%)	Recall (%)	F1 Score (%)	Observation
0.3	94.52	94.91	93.05	94.76	94.30	Better overlap, slightly reduced precision
0.5	94.93	94.61	95.25	95.76	95.65	Best overall balance, stable convergence
0.7	94.10	94.12	95.70	92.85	94.25	Faster convergence, weaker boundary segmentation

The optimal setting ($\alpha = 0.5$) provided the best trade-off between classification accuracy and segmentation overlap

Bold represents proposed values

4.4 Computation Time and Space Requirements

Table 11 compares the computation time and space requirements of the proposed MobileNetV2-UNet with ResNet-UNet and DenseNet-UNet. Both ResNet- and DenseNet-based models required considerably longer training times (~ 4.8 – 5.2 hours per fold) and slower inference speeds (25 – 32 ms per slice), in addition to occupying large model sizes (95 – 110 MB). By contrast, the proposed MobileNetV2-UNet completed training in ~ 2.3 hours, achieved near real-time inference at 12 – 15 ms per slice, and required only 18 MB of storage.

This efficiency stems from the lightweight MobileNetV2 backbone, which leverages depthwise separable convolutions and inverted residuals to reduce computational complexity without sacrificing accuracy. The results demonstrate that the proposed method achieves the best trade-off between performance and resource usage, making it well-suited for clinical environments where both accuracy and efficiency are critical.

As summarized in Table 11, the proposed MobileNetV2-UNet achieves substantially faster training and inference times with a markedly smaller model size compared to ResNet- and DenseNet-based architectures, underscoring its efficiency advantage for real-world clinical deployment.

4.5 Comparison with State-of-the-Art Models

The proposed MobileNetV2-UNet is an improvement of several existing medical models in the classification and segmentation of degenerative diseases of the lumbar spine, as illustrated in Table 12. The proposed model provides high-resolution segmentation, scalability for real-time clinical application, and the fact that the proposed model can be trained on datasets from different patients. This approach is especially essential when compared to the other existing procedures. Compared with prior studies, which often report strong performance on internal or competition datasets, our study shares the limitation of relying exclusively on the RSNA 2024 dataset. While this provides a standardized benchmark for fair comparison, it does not fully reflect the heterogeneity of real-world clinical imaging. This gap underscores the importance of testing the model on independent datasets not used for training or validation, a limitation that we explicitly acknowledge and plan to address in future work. While the comparison in Table 8 provides useful context, it is important to note that the methods listed have been evaluated on different datasets. For a more meaningful comparison, future work will focus on re-implementing these baseline models on the RSNA 2024 dataset to provide direct performance benchmarks across the same data. This will enable a fairer and more standardized comparison of the proposed method with state-of-the-art models.

For example, Jamaludin et al. [5] introduced SpineNet, which had a level of accuracy of 88.6% on MRI data concentrating on disk herniation and stenosis. Nevertheless, the absence of high-resolution segmentation, relatively low granularity, and the capacity to generalize these findings are apparent research limitations that the proposed MobileNetV2-UNet model resolves. The proposed model not only expands the scope of spinal conditions that it categorizes, but also has the ability to be implemented in real-time on the client side, which is not available in previous models. Moreover, the complexity of implementation of the proposed model is well within the capacity of typical mobile phones.

Table 11 Comparison of computation time and space requirements for different encoder backbones

Model	Training time (per fold, 50 epochs)	Inference time (per slice)	Model size	Observation
ResNet-UNet [?]	~ 4.8 hours	25 – 30 ms	~ 95 MB	High accuracy but computationally heavy
DenseNet-UNet [?]	~ 5.2 hours	28 – 32 ms	~ 110 MB	Strong performance, large memory footprint
Proposed (MobileNetV2-UNet)	~ 2.3 hours	12 – 15 ms	18 MB	Best trade-off between accuracy and efficiency
The proposed MobileNetV2-UNet achieves the most favorable balance of accuracy, speed, and model size, enabling efficient deployment in clinical practice				

Table 12 Comparison of the proposed model with state-of-the-art models

Study	Year	Methodology	Dataset	Classes	Accuracy (%)
Proposed model	2024	MobileNetV2-UNet	RSNA 2024	Left/Right Neural Foraminal Narrowing, Left/Right Subarticular Stenosis, Spinal Canal Stenosis	94.93%
Germann et al. [6]	2023	CNN	MRI vertebral body dataset	Vertebral insufficiency fracture detection	92.4%
Ma et al. [28]	2024	Vision transformer (ViT)	MRI of various medical conditions	Segmentation of spine-related structures	92.1%
Su et al. [21]	2022	ResNet-50	Lumbar spine MRI images	Grading disc herniation, stenosis, nerve root compression	91.5%
Song et al. [25]	2023	Mask R-CNN	PA X-ray images of the spine	Sagittal spine segmentation	91.0%
Zhou et al. [4]	2019	CNN	Lumbar spine MRI dataset	Degenerative disease detection and classification	90.1%
Siddique et al. [12]	2021	U-Net variants	MRI spine segmentation	Segmentation of degenerative changes	90.0%
Almansour et al. [26]	2022	Deep learning	Accelerated spine MRI dataset	MRI reconstruction	89.8%
Saravi et al. [10]	2023	U-Net	Lumbar spine MRI dataset	Spinal stenosis diagnosis	89.2%
Jamaludin et al. [5]	2017	CNN	MRI of lumbar spine	Disk herniation, stenosis, facet arthropathy	88.6%
Bharadwaj et al. [8]	2023	V-Net, CNN	Axial MRI images of the lumbar spine	Spinal stenosis and facet arthropathy classification	87.5%
Hallinan et al. [11]	2021	CNN	MRI lumbar spine dataset	Central canal stenosis, neural foraminal narrowing	85.0%

Similarly, Zhou et al. [4] proposed a deep learning model to perform an initial identification of degenerative disease at 90.1% accuracy. This model used lumbar MRI datasets. However, it had no interpretability and was not designed for real-time hospital applications. The MobileNetV2-UNet model, on the other hand, considers both accuracy and efficiency in computation and, as such, can be deployed in clinical settings without significantly reducing the level of performance, as revealed by the 94.93% accuracy achieved from the RSNA 2024 dataset.

Other methods, including Hallinan et al. [12] and Germann et al. [6], achieved an accuracy of 85.0% and 92.4%, respectively, when the researchers applied CNN-based approaches. However, these models did not emphasize clinical use in real-time and generalization across clinical contexts and data. The proposed model overcomes these shortcomings with the help of the MobileNetV2 backbone and employs higher interpretability methods, such as Grad-CAM, to address the problem to make a more adaptive solution for various medical settings and provide clear visualizations that aid in clinical decision-making and increase its credibility.

The present MobileNetV2-UNet model differs from models of Bharadwaj et al. [8] and Saravi et al. [10] on interpretability and segmentation of specific spinal diseases like left foraminal narrowing, right foraminal narrowing and subarticular stenosis and so forth. The expanded number of values, which are assigned to particular classes, along with the high accuracy of the model, makes the presented approach more suitable for the diagnostics of lumbar spine conditions.

However, the last model by Ma et al. [28] used a Vision Transformer (ViT) for spine segmentation with 92.1% accuracy, as shown in Table 12. Although this model was beneficial, it needed to incorporate diagnostic and classification tools. It needed to be optimized for real-time deployment, one of the unique features of the proposed MobileNetV2-UNet model. The proposed model outperforms most of the related models. At the same time, it

ensures that minimal computational resources are needed, making it ideal for real-time clinical applications, especially in resource-constrained settings.

Therefore, the proposed MobileNetV2-UNet model provides reliable high-resolution segmentation, operates in near real-time, can be generalized for other datasets, and, most importantly, is interpretable, making it superior to existing state-of-the-art methods. It also closes the gaps of previous models by providing a systemic and expandable framework for lumbar spine degenerative disease classification, which aligns with the contributions presented in the attached article. This model is suited to replace traditional clinical diagnostics and direction due to the accuracy and speed of the distance results, and it is easy to interpret in a healthcare setting.

5 Conclusion

This study developed an efficient and scalable deep learning framework, combining MobileNetV2 and UNet, for the automated classification and segmentation of lumbar spine degenerative diseases. By leveraging the lightweight MobileNetV2 for feature extraction and the robust UNet architecture for segmentation, the model demonstrated strong performance, achieving 94.93% accuracy and 94.61% Dice Coefficient on the RSNA 2024 dataset. In summary, the proposed MobileNetV2-UNet not only outperforms several state-of-the-art methods in terms of segmentation accuracy but also offers significantly reduced computation time and memory footprint. With a training time of 2.3 h per fold, real-time inference capability (12–15 ms per slice), and a compact model size of 18 MB, the framework achieves a practical balance between performance and efficiency. These strengths position it as a clinically viable solution for large-scale deployment, especially in settings where computational resources are limited. The use of depthwise separable convolutions ensured the model was computationally efficient, making it suitable for real-time clinical applications on resource-constrained devices. The integration of skip connections preserved spatial details, enhancing segmentation accuracy, while Grad-CAM provided interpretability, building clinician trust in the model's predictions. The model successfully classified and segmented multiple lumbar spine conditions, including neural foraminal narrowing, subarticular stenosis, and spinal canal stenosis. It can operate with different types of data, which can enhance diagnostic procedures in healthcare facilities, as it reduces the burden on radiologists and improves diagnostic capabilities.

Other approaches to explainable AI, such as attention processes, might help improve the model's interpretability, which could be addressed in future studies. Our future research will validate the MobileNetV2-UNet model using MRI datasets from multiple hospitals, covering diverse scanner types, settings, and patient groups to ensure it works well in varied clinical settings. We'll explore domain adaptation and transfer learning to boost the model's ability to handle different environments, making it more practical for real-world medical use. Additionally, we plan to add severity grading by analyzing segmentation masks for features like canal area or foraminal width, linking them to clinical scales like Schizas or Pfirrmann for better treatment alignment. To strengthen robustness, we'll test the model on independent, multi-institutional datasets and compare it directly with top existing models on the RSNA 2024 dataset, ensuring fair and clear performance evaluations.

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Data Availability The appropriate author can make the datasets used in this study available upon request.

Declarations

Competing Interests The authors declare no competing interests.

Ethical Approval and Consent to Participate This study is based on the Kaggle competition RSNA 2024 Lumbar Spine Degenerative Classification. This competition aims to create models that can aid in the detection and classification of degenerative spine conditions using lumbar spine MR images. The RSNA team obtained the consent of the participants.

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References

1. Mamisch, N., Brumann, M., Hodler, J., Held, U., Brunner, F., Steurer, J.: Radiologic criteria for the diagnosis of spinal stenosis: results of a delphi survey. *Radiol.* **264**(1), 174–179 (2012)
2. Andreisek, G., Deyo, R.A., Jarvik, J.G., Porchet, F., Winklhofer, S.F., Steurer, J., Group, L.W.: Consensus conference on core radiological parameters to describe lumbar stenosis—an initiative for structured reporting. *Eur. Radiol.* **24**, 3224–3232 (2014)
3. Jamaludin, A., Lootus, M., Kadir, T., Zisserman, A., Urban, J., Battié, M.C., Fairbank, J., McCall, I., Consortium, G.: Issls prize in bioengineering science: automation of reading of radiological features from magnetic resonance images (mris) of the lumbar spine without human intervention is comparable with an expert radiologist. *Eur. Spine J.* **26**(2017), 1374–1383 (2017)
4. Zhou, Y., Liu, Y., Chen, Q., Gu, G., Sui, X.: Automatic lumbar mri detection and identification based on deep learning. *J. Digit. Imaging* **32**, 513–520 (2019)
5. Jamaludin, A., Kadir, T., Zisserman, A.: Spinenet: automated classification and evidence visualization in spinal mris. *Med. Image Anal.* **41**, 63–73 (2017)
6. Germann, C., Meyer, A.N., Staib, M., Sutter, R., Fritz, B.: Performance of a deep convolutional neural network for mri-based vertebral body measurements and insufficiency fracture detection. *Eur. Radiol.* **33**(5), 3188–3199 (2023)
7. Fritz, B., Paul, H.Y., Kijowski, R., Fritz, J.: Radiomics and deep learning for disease detection in musculoskeletal radiology: an overview of novel mri-and ct-based approaches. *Invest. Radiol.* **58**(1), 3–13 (2023)
8. Bharadwaj, U.U., Christine, M., Li, S., Chou, D., Pedoia, V., Link, T.M., Chin, C.T., Majumdar, S.: Deep learning for automated, interpretable classification of lumbar spinal stenosis and facet arthropathy from axial mri. *Eur. Radiol.* **33**(5), 3435–3443 (2023)
9. Ishimoto, Y., Jamaludin, A., Cooper, C., Walker-Bone, K., Yamada, H., Hashizume, H., Oka, H., Tanaka, S., Yoshimura, N., Yoshida, M., et al.: Could automated machine-learned mri grading aid epidemiological studies of lumbar spinal stenosis? validation within the wakayama spine study. *BMC Musculoskelet. Disord.* **21**, 1–6 (2020)
10. Saravi, B., Zink, A., Ülkümen, S., Couillard-Despres, S., Wollborn, J., Lang, G., Hassel, F.: Automated detection and measurement of dural sack cross-sectional area in lumbar spine mri using deep learning. *Bioeng.* **10**(9), 1072 (2023)
11. Hallinan, J.T.P.D., Zhu, L., Yang, K., Makmur, A., Algazwi, D.A.R., Thian, Y.L., Lau, S., Choo, Y.S., Eide, S.E., Yap, Q.V., et al.: Deep learning model for automated detection and classification of central canal, lateral recess, and neural foraminal stenosis at lumbar spine mri. *Radiol.* **300**(1), 130–138 (2021)
12. Siddique, N., Paheding, S., Elkin, C.P., Devabhaktuni, V.: U-net and its variants for medical image segmentation: a review of theory and applications. *IEEE Access* **9**, 82031–82057 (2021)
13. Lurie, J., Tomkins-Lane, C.: Management of lumbar spinal stenosis. *Bmj* **352**, h6234 (2016)

14. Katz, J.N., Harris, M.B.: Lumbar spinal stenosis. *N. Engl. J. Med.* **358**(8), 818–825 (2008)
15. Thalgott, J.S., Albert, T.J., Vaccaro, A.R., Aprill, C.N., Giuffre, J.M., Drake, J.S., Henke, J.P.: A new classification system for degenerative disc disease of the lumbar spine based on magnetic resonance imaging, provocative discography, plain radiographs and anatomic considerations. *The Spine J.* **4**(6), S167–S172 (2004)
16. Stephens, K.: Rsna launches lumbar spine degenerative classification ai challenge. *AXIS Imaging News* (2024)
17. Galbusera, F., Niemeyer, F., Wilke, H.-J., Bassani, T., Casaroli, G., Anania, C., Costa, F., Brayda-Bruno, M., Sconfienza, L.M.: Fully automated radiological analysis of spinal disorders and deformities: a deep learning approach. *Eur. Spine J.* **28**, 951–960 (2019)
18. Pan, S., Liu, X., Xie, N., Chong, Y.: Eg-transunet: a transformer-based u-net with enhanced and guided models for biomedical image segmentation. *BMC Bioinf.* **24**(1), 85 (2023)
19. Shamshad, F., Khan, S., Zamir, S.W., Khan, M.H., Hayat, M., Khan, F.S., Fu, H.: Transformers in medical imaging: a survey. *Med. Image Anal.* **88**, 102802 (2023)
20. Antun, V., Renna, F., Poon, C., Adcock, B., Hansen, A.C.: On instabilities of deep learning in image reconstruction and the potential costs of ai. *Proc. Natl. Acad. Sci.* **117**(48), 30088–30095 (2020)
21. Su, Z.-H., Liu, J., Yang, M.-S., Chen, Z.-Y., You, K., Shen, J., Huang, C.-J., Zhao, Q.-H., Liu, E.-Q., Zhao, L., et al.: Automatic grading of disc herniation, central canal stenosis and nerve roots compression in lumbar magnetic resonance image diagnosis. *Front. Endocrinol.* **13**, 890371 (2022)
22. Park, J., Yang, J., Park, S., Kim, J.: Deep learning-based approaches for classifying foraminal stenosis using cervical spine radiographs. *Electronics* **12**(1), 195 (2022)
23. Yu, J., Qin, J., Xiang, J., He, X., Zhang, W., Zhao, W.: Trans-uneter: a new decoder of transunet for medical image segmentation. In: 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), IEEE, pp. 2338–2341 (2023)
24. Yeh, Y.-C., Weng, C.-H., Huang, Y.-J., Fu, C.-J., Tsai, T.-T., Yeh, C.-Y.: Deep learning approach for automatic landmark detection and alignment analysis in whole-spine lateral radiographs. *Sci. Rep.* **11**(1), 7618 (2021)
25. Song, S.-Y., Seo, M.-S., Kim, C.-W., Kim, Y.-H., Yoo, B.-C., Choi, H.-J., Seo, S.-H., Kang, S.-W., Song, M.-G., Nam, D.-C., et al.: AI-driven segmentation and automated analysis of the whole sagittal spine from x-ray images for spinopelvic parameter evaluation. *Bioengineering* **10**(10), 1229 (2023)
26. Almansour, H., Herrmann, J., Gassenmaier, S., Afat, S., Jacoby, J., Koerzdoerfer, G., Nickel, D., Mostapha, M., Nadar, M., Othman, A.E.: Deep learning reconstruction for accelerated spine mri: prospective analysis of interchangeability. *Radiology* **306**(3), e212922 (2022)
27. Payne, D.L., Xu, X., Faraji, F., John, K., Pradas, K.F., Bernard, V.V., Bangiyev, L., Prasanna, P.: Automated detection of cervical spinal stenosis and cord compression via vision transformer and rules-based classification. *Am. J. Neuroradiol.* **45**(4), 432–438 (2024)
28. Ma, J., He, Y., Li, F., Han, L., You, C., Wang, B.: Segment anything in medical images. *Nat. Commun.* **15**(1), 654 (2024)
29. Pradhan, P.K., Das, A., Kumar, A., Baruah, U., Sen, B., Ghosal, P.: Swinsight: a hierarchical vision transformer using shifted windows to leverage aerial image classification. *Multimed. Tools and Appl.* **83**(39), 86457–86478 (2024)
30. Li, X., Hong, Y., Xu, Y., Hu, M.: Verformer: vertebrae-aware transformer for automatic spine segmentation from ct images. *Diagnostics* **14**(17), 1859 (2024)
31. Lin, D.J., Johnson, P.M., Knoll, F., Lui, Y.W.: Artificial intelligence for mr image reconstruction: an overview for clinicians. *J. Magn. Reson. Imaging* **53**(4), 1015–1028 (2021)
32. Hammernik, K., Klatzer, T., Kobler, E., Recht, M.P., Sodickson, D.K., Pock, T., Knoll, F.: Learning a variational network for reconstruction of accelerated mri data. *Magn. Reson. Med.* **79**(6), 3055–3071 (2018)
33. Zhang, Z., Wu, H., Zhao, H., Shi, Y., Wang, J., Bai, H., Sun, B.: A novel deep learning model for medical image segmentation with convolutional neural network and transformer. *Interdiscip. Sci. Comput. Life Sci.* **15**(4), 663–677 (2023)
34. Loder, R.T., Urquhart, A., Steen, H.U., Graziano, G., Hensinger, R., Schlesinger, A., Schork, M., Shyr, Y.: Variability in cobb angle measurements in children with congenital scoliosis. *The J. Bone & Jt. Surg. Br.* **77**(5), 768–770 (1995)
35. Constant, C., Aubin, C.-E., Kremers, H.M., Garcia, D.V.V., Wyles, C.C., Rouzrokh, P., Larson, A.N.: The use of deep learning in medical imaging to improve spine care: a scoping review of current literature and clinical applications. *North Am. Spine Soc. J. (NASSJ)* **15**, 100236 (2023)
36. Tong, N., Gou, S., Yang, Y., Liu, B., Bai, Y., Liu, J., Ding, T.: Fully automatic fine-grained grading of lumbar intervertebral disc degeneration using regional feature recalibration. *IEEE J. Biomed. Health Inf.* **28**(5), 3042–3054 (2024)
37. Rujeedawa, T., Mowforth, O.D., Davies, B.M., Yang, C., Nouri, A., Francis, J.J., Aarabi, B., Kwon, B.K., Harrop, J., Wilson, J.R., et al.: Degenerative thoracic myelopathy: a scoping review of epidemiology, genetics, and pathogenesis. *Global Spine J.* **14**(5), 1664–1677 (2024)
38. Cawley, D.T., Simpkin, A., Abraham, E., Doyle, T., Elsheikh, N., Fallon, J., Habash, M., Phua, R.J., Langille, J., Matini, E., et al.: Intradiscal vacuum phenomenon matches lumbar spine degeneration patterns in an ageing population. *Euro. Spine J.* **33**, 1–8 (2024)

39. Rsna 2024 lumbar spine degenerative classification, accessed: 2024-09-14 (2024). <https://www.kaggle.com/competition/s/rsna-2024-lumbar-spine-degenerative-classification>
40. Goodfellow, I., Bengio, Y., Courville, A.: Deep Learning. MIT Press (2016)
41. Krizhevsky, A., Sutskever, I., Hinton, G.E.: Imagenet classification with deep convolutional neural networks. *Commun. ACM* **60**(6), 84–90 (2017)
42. Simonyan, K., Zisserman, A.: Very deep convolutional networks for large-scale image recognition, arXiv preprint [arXiv:1409.1556](https://arxiv.org/abs/1409.1556) (2014)
43. LeCun, Y., Bottou, L., Orr, G.B., Müller, K.-R.: Efficient backprop. In: *Neural networks: Tricks of the trade*, pp. 9–50. Springer (2002)
44. Ioffe, S., Szegedy, C.: Batch normalization: accelerating deep network training by reducing internal covariate shift. In: *Proceedings of the 32nd International Conference on Machine Learning (ICML-15)*, pp. 448–456 (2015)
45. Bishop, C.M., Nasrabadi, N.M.: *Pattern Recognition and Machine Learning*, vol. 4. Springer, New York (2006)
46. Rashid, J., Khan, I., Abbasi, I.A., Saeed, M.R., Saddique, M., Abbas, M.: A hybrid deep learning approach to classify the plant leaf species. *CMC-Comput. Mater. Contin.* **76**(3), 3897–3920 (2023)
47. Chu, H.-H., Saeed, M.R., Rashid, J., Mehmood, M.T., Ahmad, I., Iqbal, R.S., Ali, G.: Deep learning method to detect the road cracks and potholes for smart cities. *Comput. Mater. Contin.* **75**(1), 1863–1881 (2023)

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